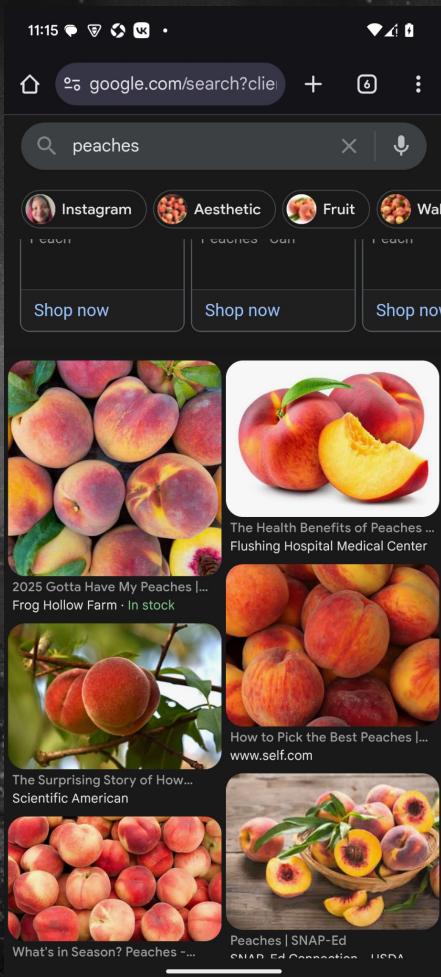
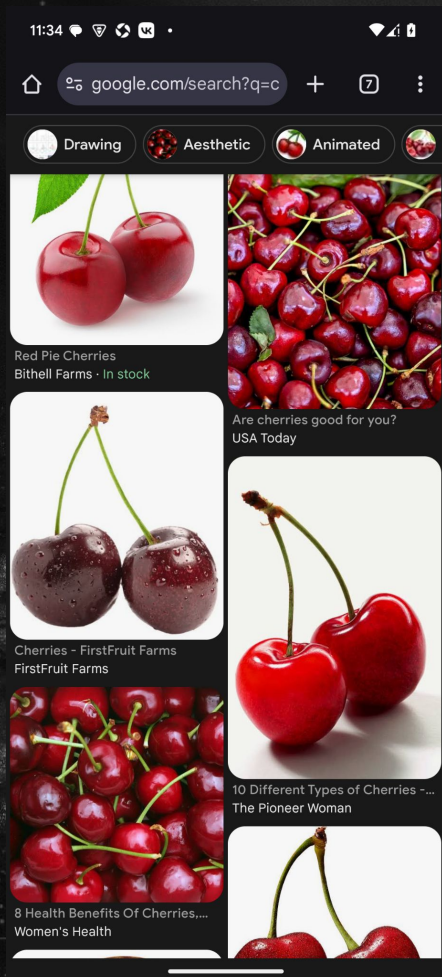


# TSPU: Russia's Firewall and Defending Against the Future of Digital Repression

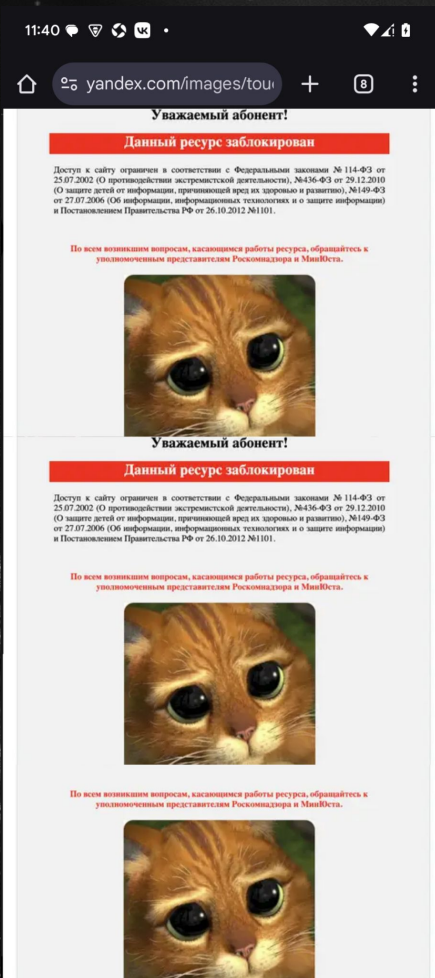


BREAKPOINTING  
BAD

Benjamin Mixon-Baca









“Hacker making a peach, cherry, and eggplant salad arrested by a drone.”





# Policies

- Федерального закона от 7 июля 2003 г. № 126-ФЗ «О связи»
- Федерального закона от 27 июля 2006 г. № 149-ФЗ «Об информации, информационных технологиях и о защите информации»
- Федеральный закон № 139-ФЗ 2012 года — дополнения «о защите детей», приведшие к созданию «Единого реестра запрещённых сайтов»;
- Федеральный закон № 187-ФЗ от 2 июля 2013 года — закон, подразумевающий возможность блокировки сайтов, содержащих нелицензионный контент, по требованию правообладателя;
- Федеральный закон № 398-ФЗ 2013 года — дополнения, связанные с блокировкой экстремистских сайтов;
- Федеральный закон № 97-ФЗ от 5 мая 2014 года — «закон о блогерах», обязывающий владельцев популярных блогов и сайтов регистрироваться в Роскомнадзоре.
- Федеральный закон № 90-ФЗ от 1 мая 2019 года — закон о «суверенном интернете», увеличение
- Federal Law of July 7, 2003 No. 126-FZ "On Communications"
- Federal Law of July 27, 2006 No. 149-FZ "On Information, Information Technologies and Information Protection"
- Federal Law No. 139-FZ of 2012 — amendments "on the **protection of children**", which led to the creation of the "Unified Register of Prohibited Sites";
- Federal Law No. 187-FZ of July 2, 2013 — a law implying the possibility of blocking sites containing unlicensed content at the request of the copyright holder;
- Federal Law No. 398-FZ of 2013 — amendments related to the blocking of extremist sites;
- Federal Law No. 97-FZ of May 5, 2014 — the "law on bloggers", obliging owners of popular blogs and sites to register with Roskomnadzor.
- Federal Law No. 90-FZ of May 1, 2019 — the law on the "sovereign Internet", increasing the stability of the Runet.

# Policies

- Федеральный закон № 90-ФЗ от 1 мая 2019 года — закон о «суверенном интернете», увеличение устойчивости Рунета.
- Federal Law No. 90-FZ of May 1, 2019 — the law on the "sovereign Internet", increasing the stability of the Runet.



# RuNet: Russia's Sovereign Internet

- Роскомнадзор (РКН)
- Rozkomnadzor (RKN)





# What's Blocked?

Navigation bar: CNN US World Politics Business Health Entertainment

Live Updates: Wimbledon women's final | Trending: Texas flooding timeline



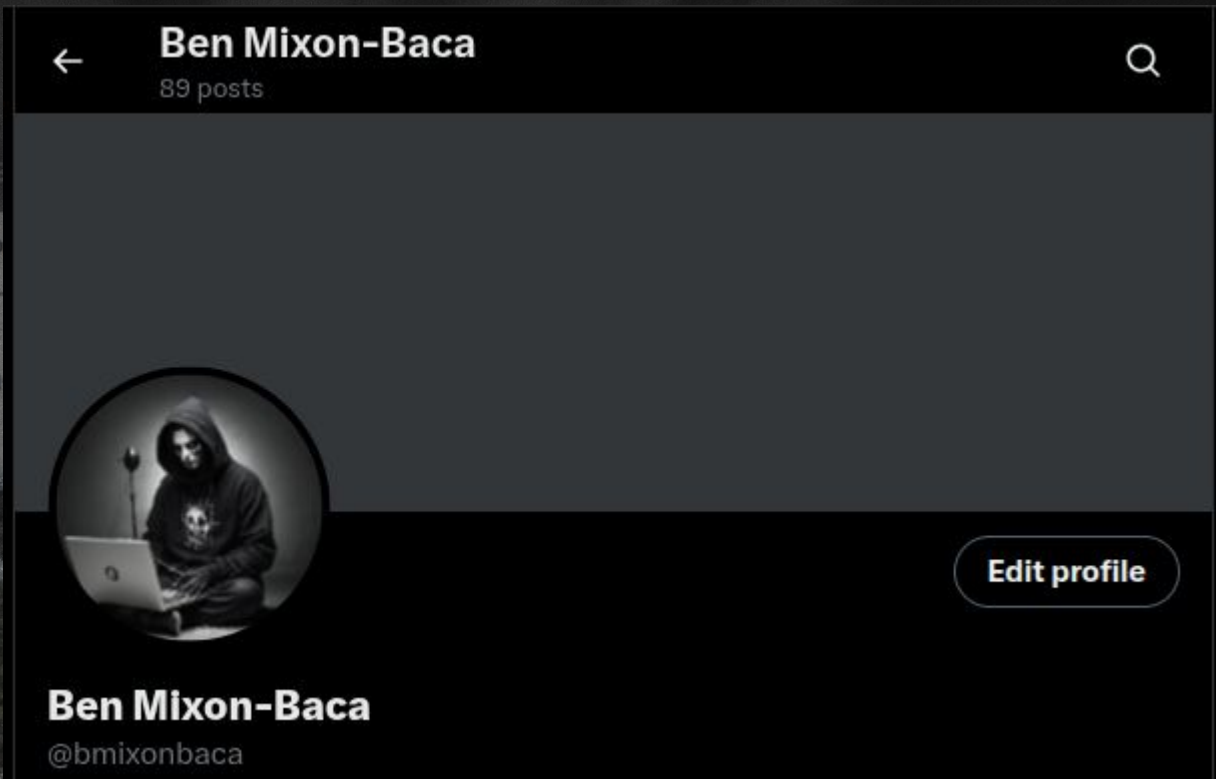
FEMA removed dozens of Camp Mystic buildings from

## Trump announce on Mexico

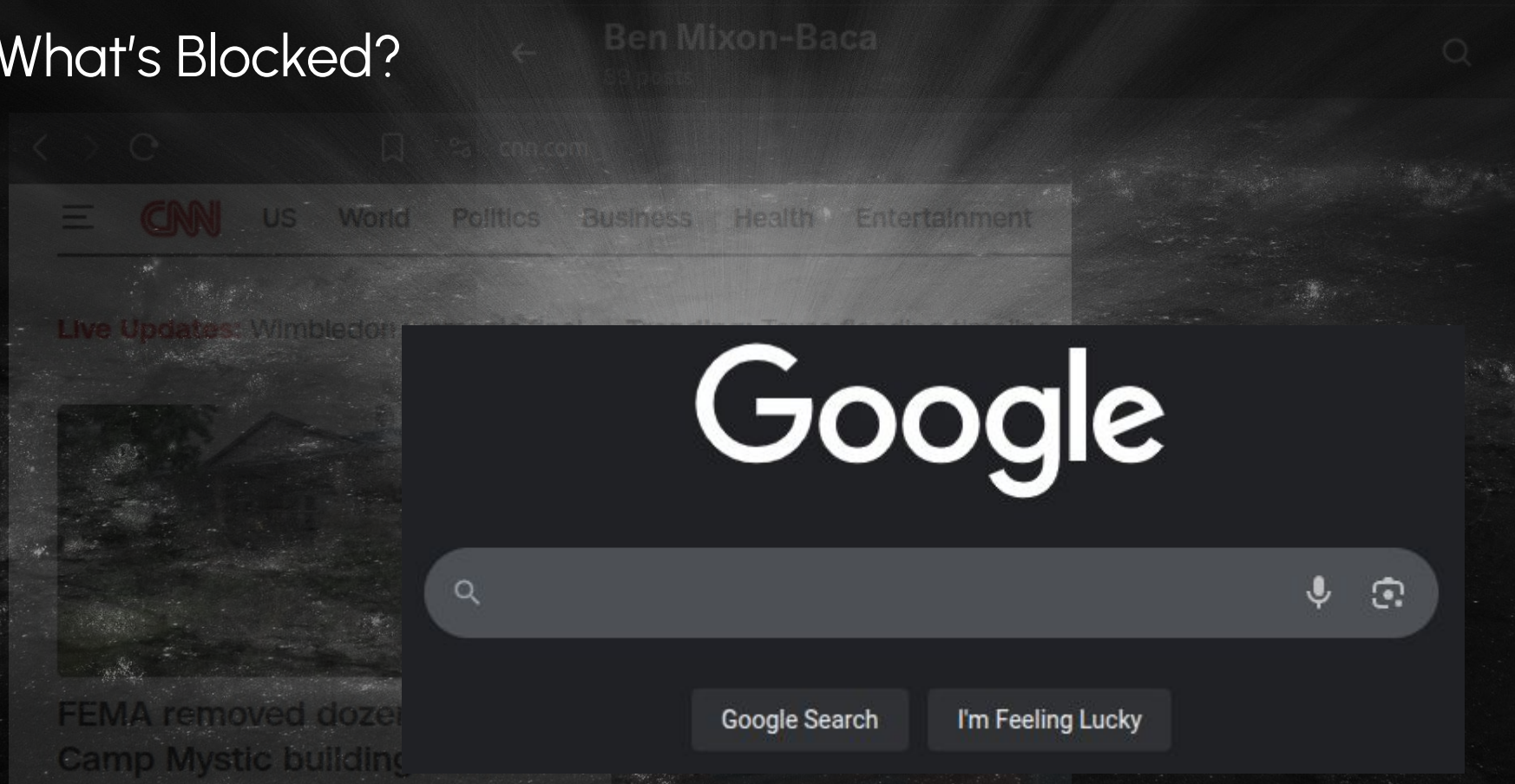




# What's Blocked?

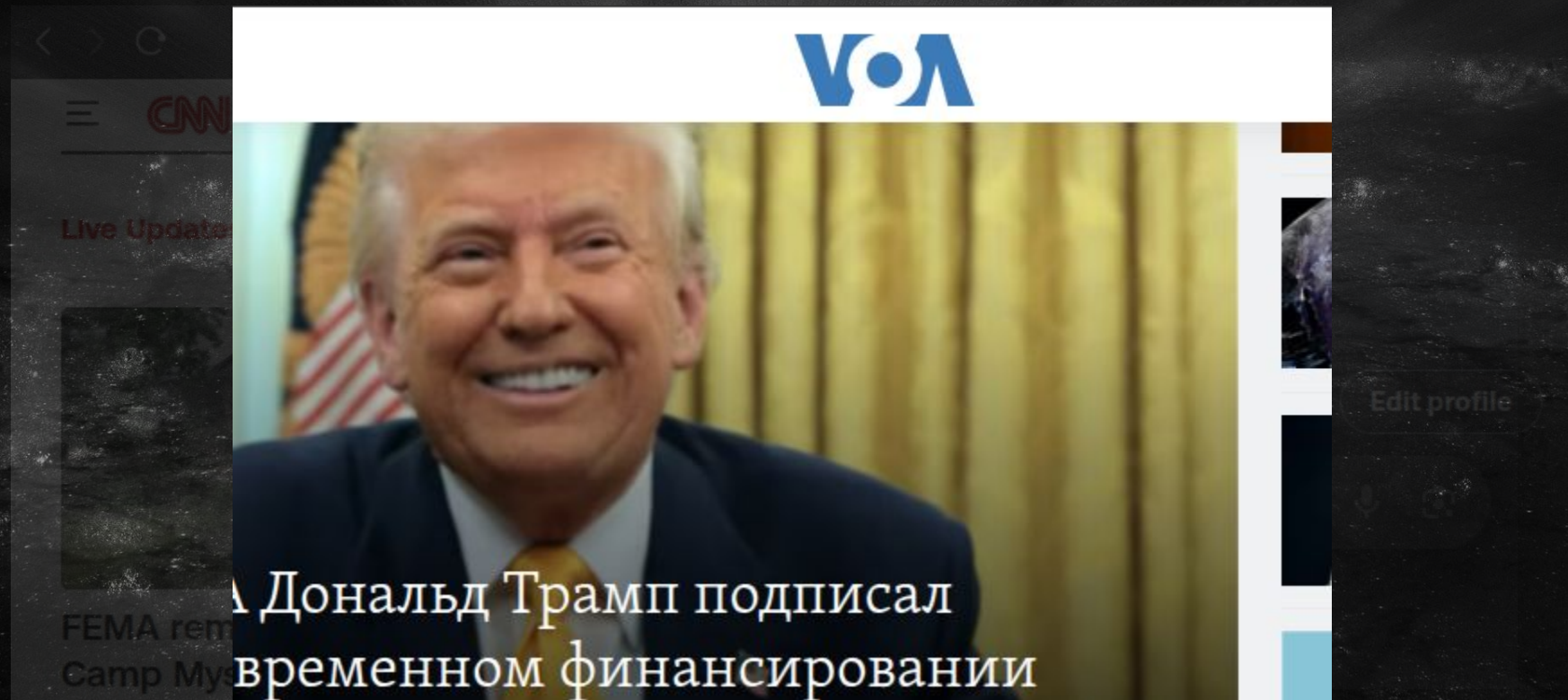


# What's Blocked?

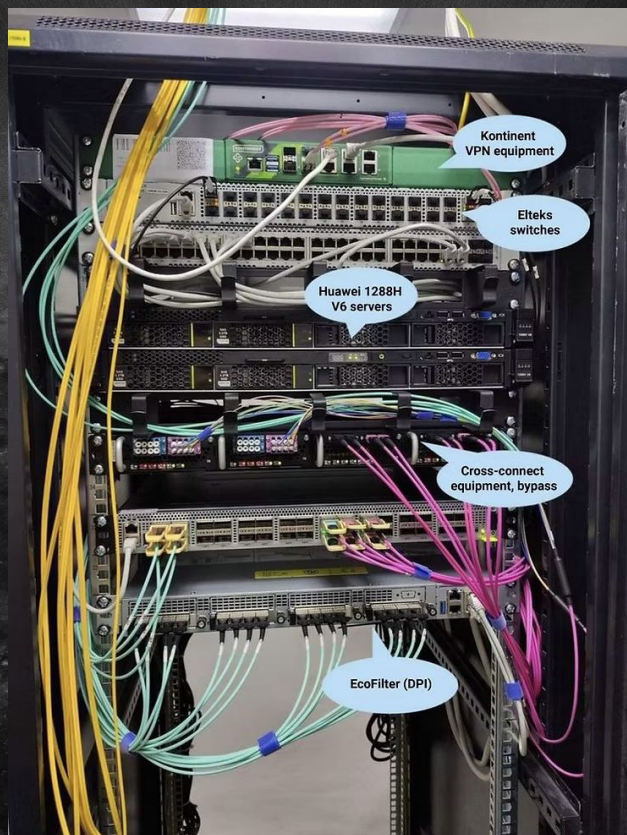




# What's Blocked?

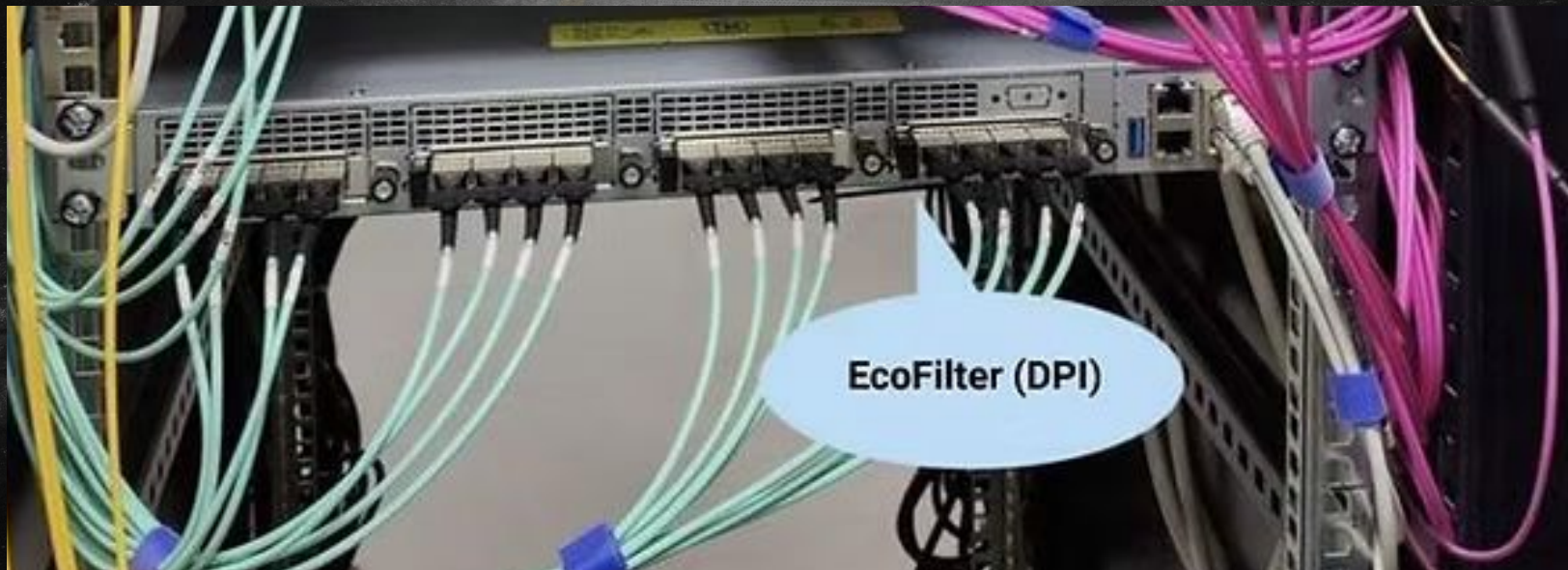


# ТСПУ: технических средств противодействия угрозам

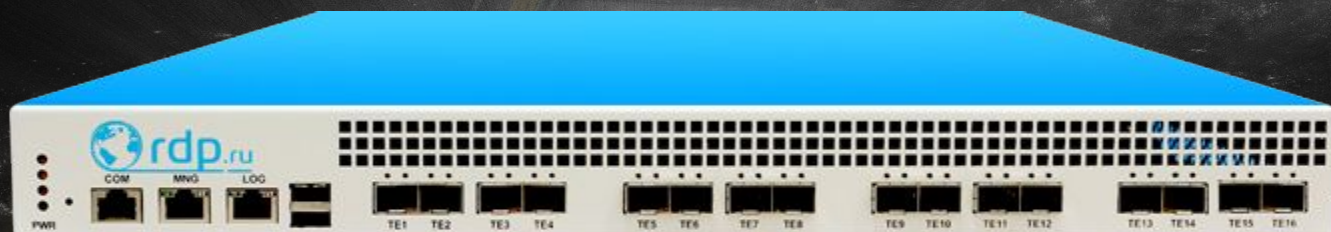




# ТСПУ: технических средств противодействия угрозам



# TSPU: Technical Means of Countering Threats

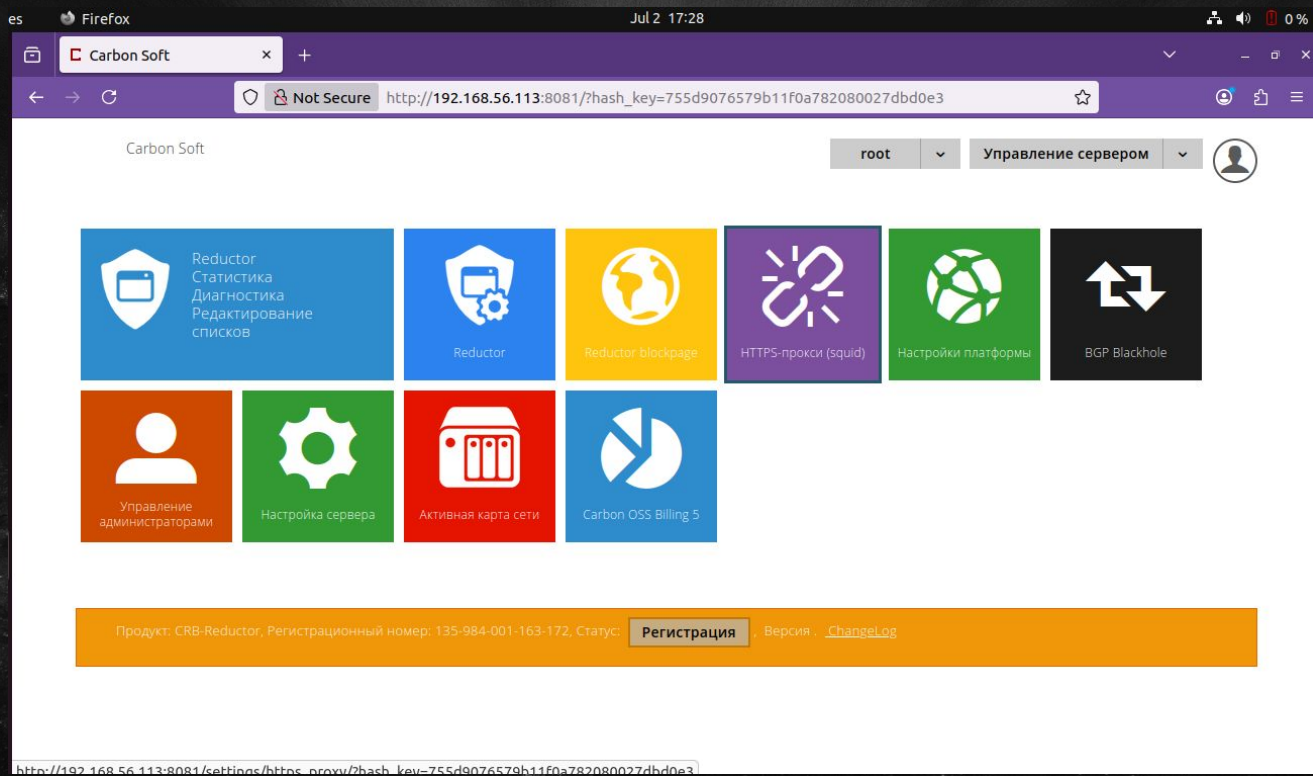




# Devices

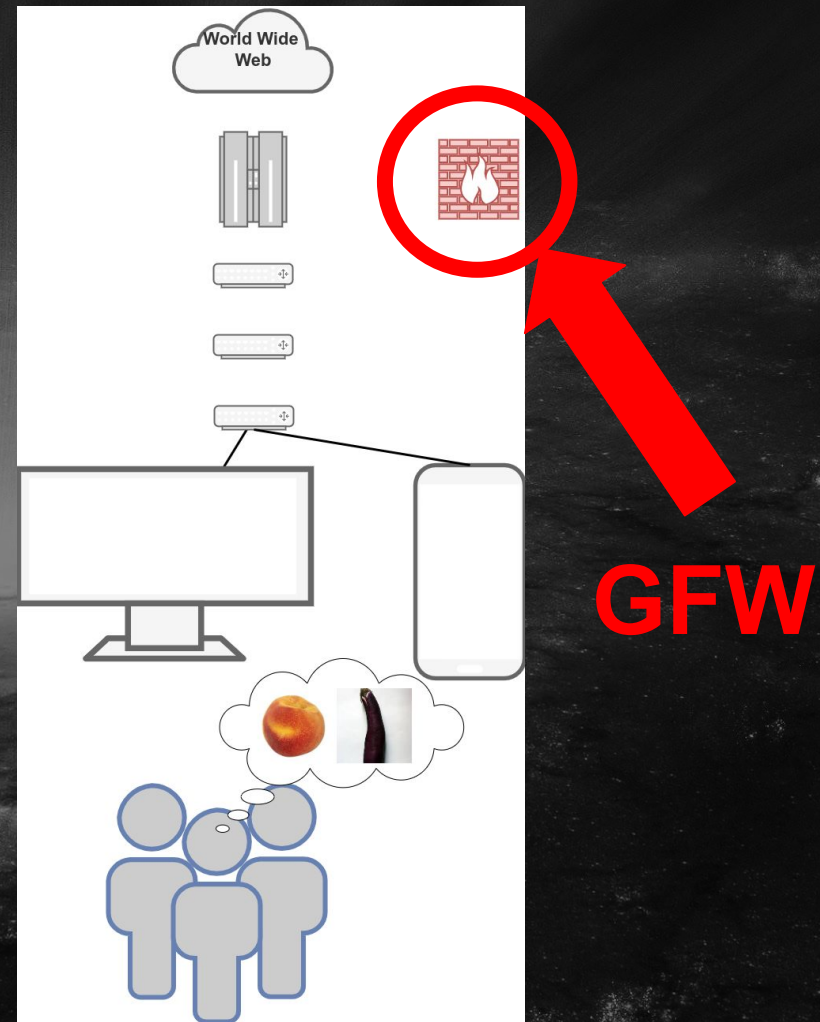
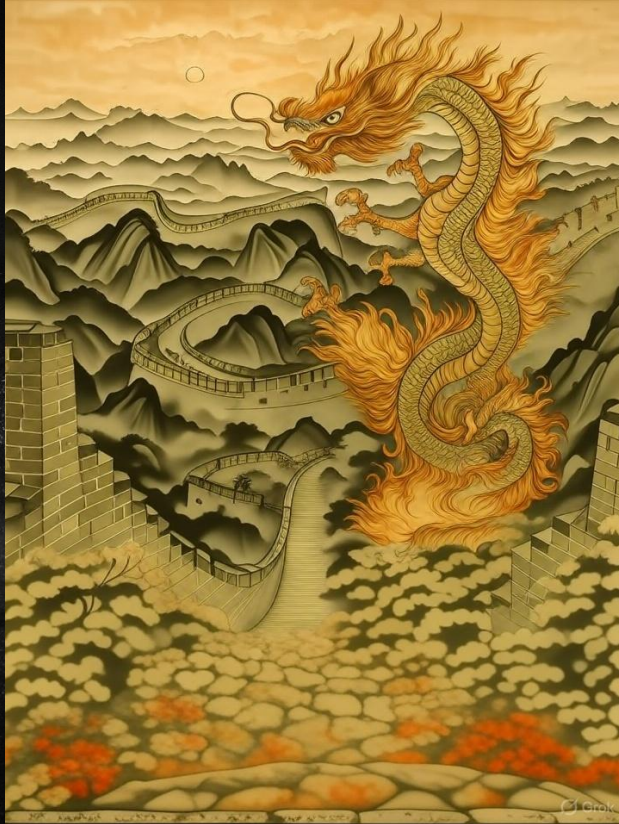
1. EcoFilter, [rdp.ru](http://rdp.ru)
2. SKAT DPI, <https://vasexperts.ru/products/skat/>
3. CPAK “Equalia”, <https://napalabs.ru/>
4. CyberFilter, <http://cyberfilter.ru/>, Linux
5. Admfilter, <http://adm-systems.com/>,
6. Zapretservice, [zapretservice.ru](http://zapretservice.ru), Linux
7. Ideco, <http://ideco.ru/>, Linux
8. Carbonfilter, <http://carbonsoft.ru/>, Linux
9. SkyDNS Zapret, Linux
10. Tixen-Blocking
11. SKAT, [vasexperts.ru](http://vasexperts.ru)
12. UBIC, q

# Other Platforms - CarbonFilter

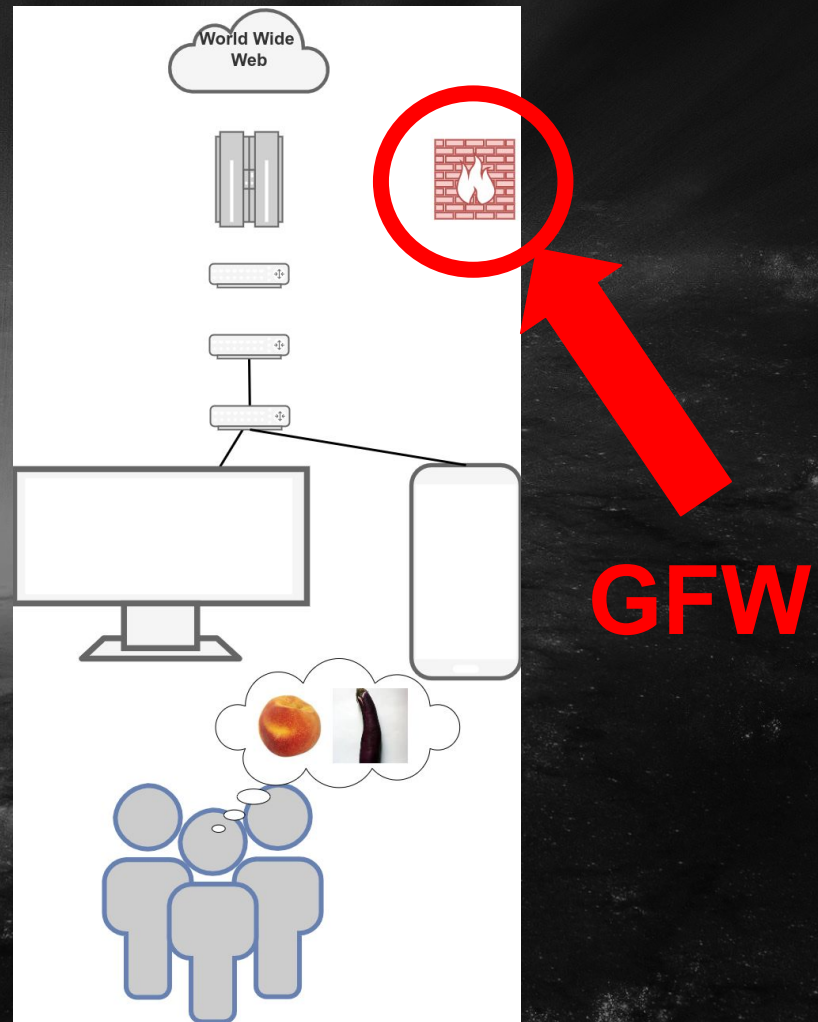




GFW est. 2005

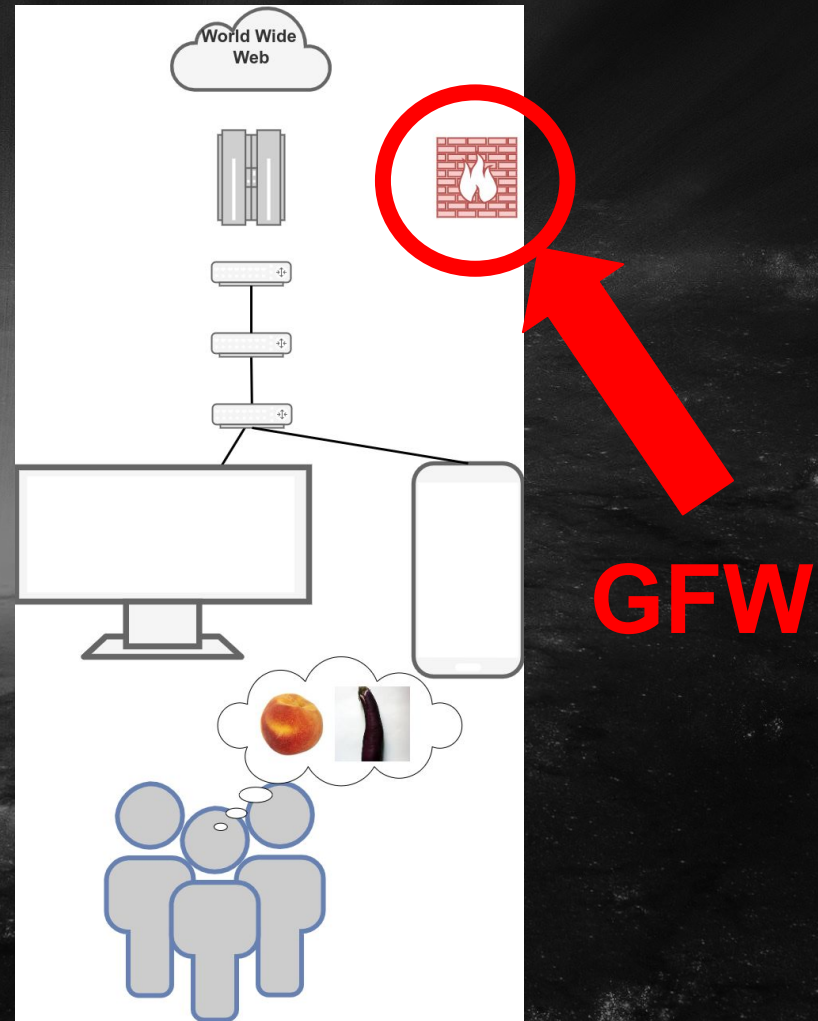
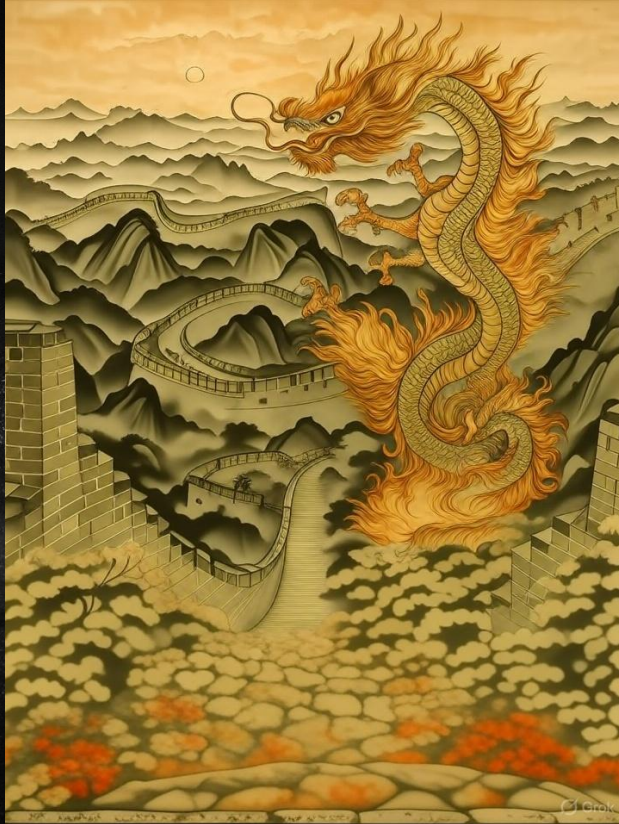


GFW est. 2005

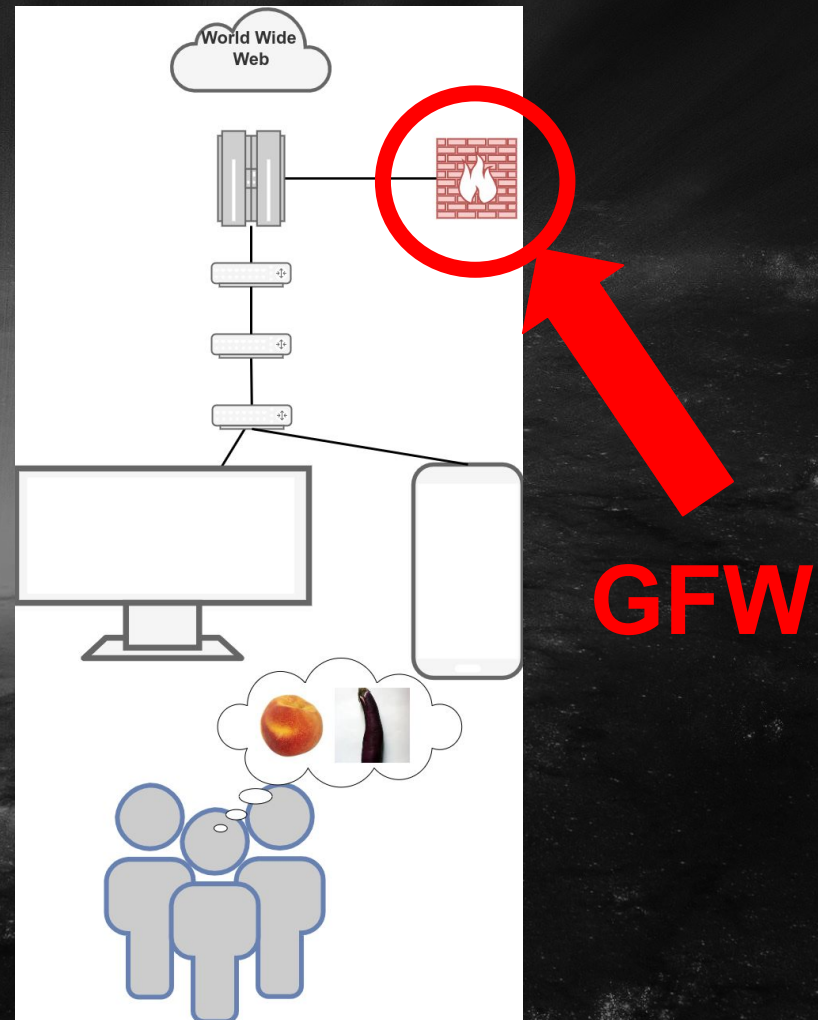




GFW est. 2005

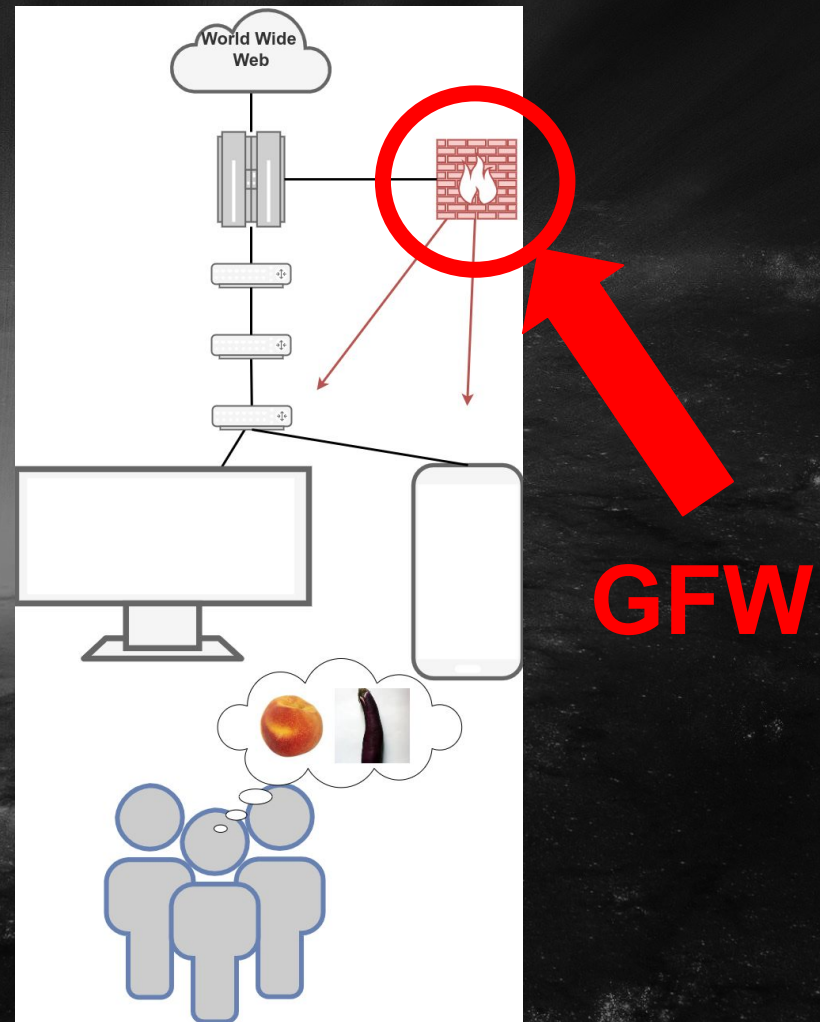
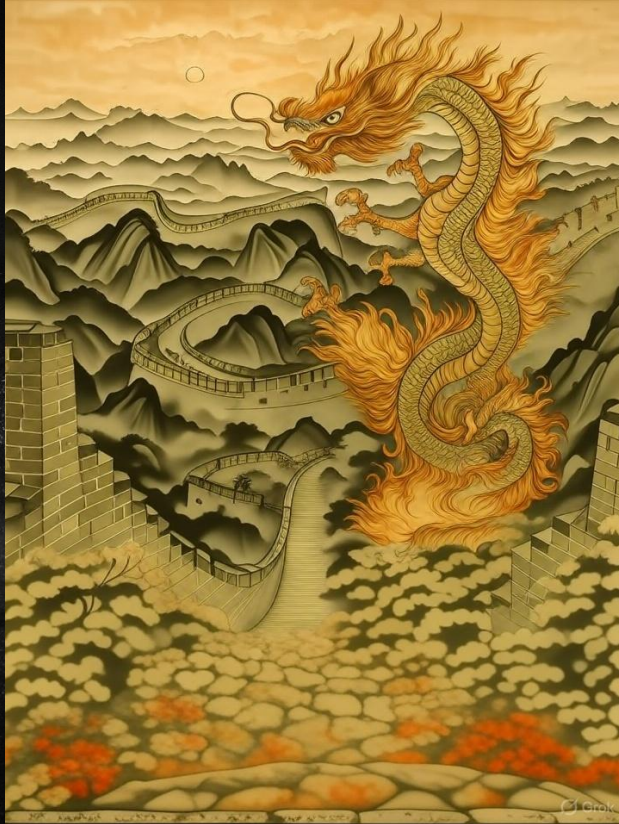


GFW est. 2005

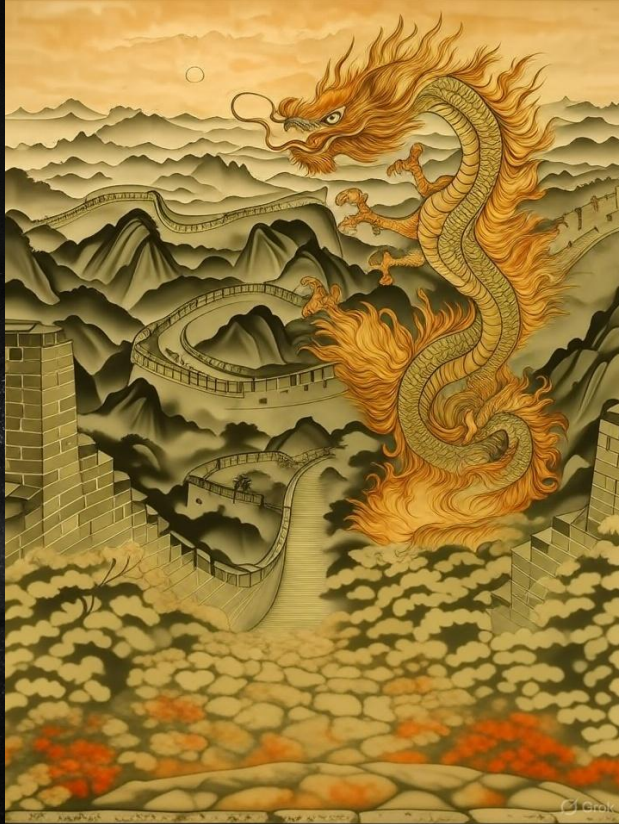




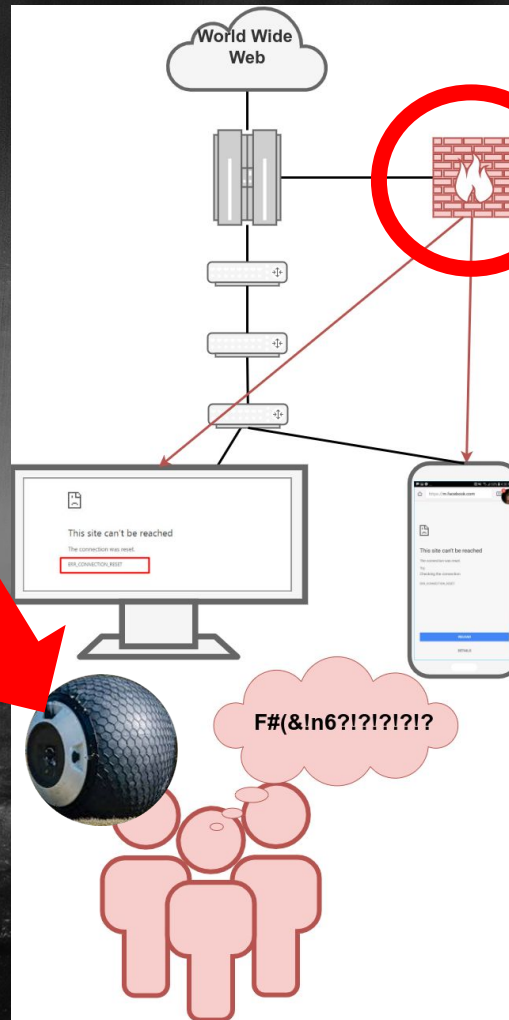
GFW est. 2005



# GFW est. 2005: RST Injection



Scary  
police  
drone

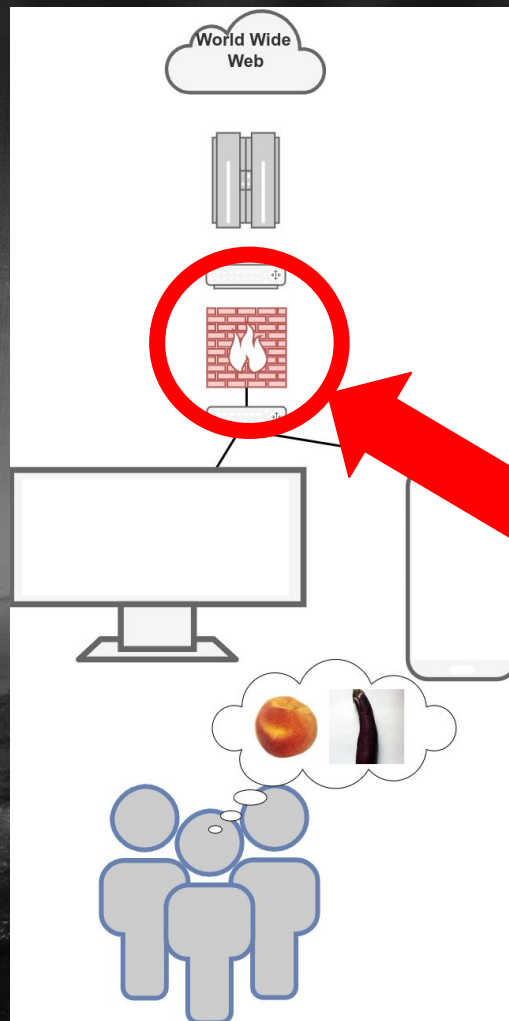




TSPU: est. 2017



TSPU: est. 2017



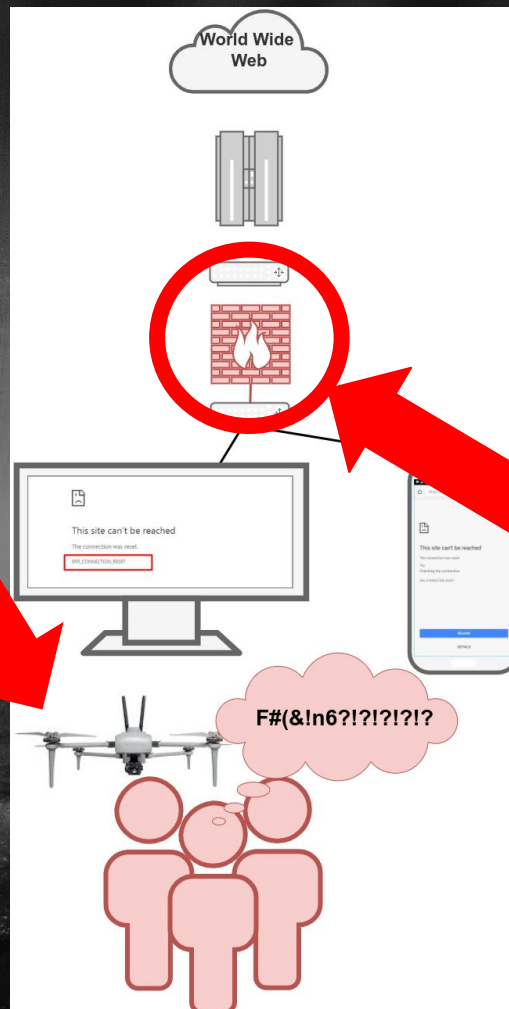
**TSPU**



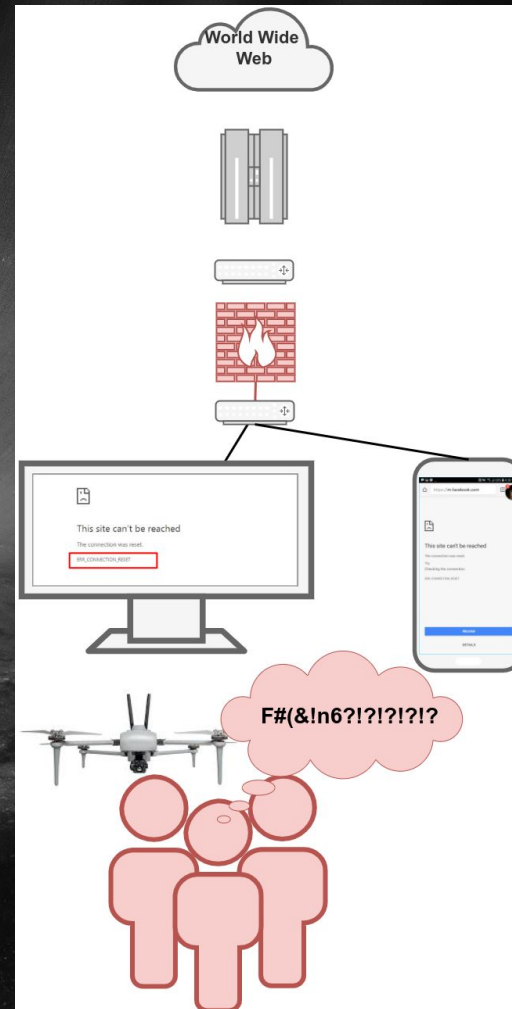
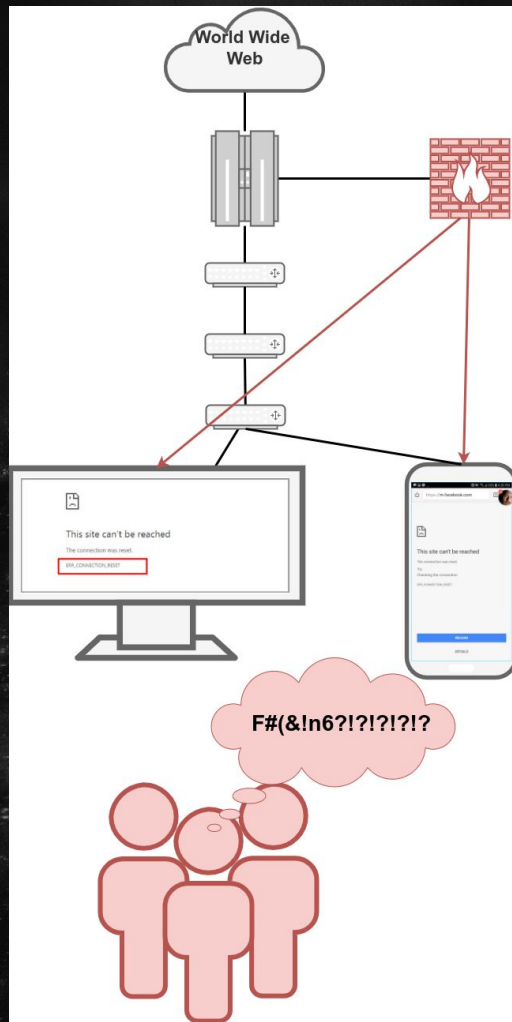
TSPU: est. 2017



Scary  
police  
drone



TSPU





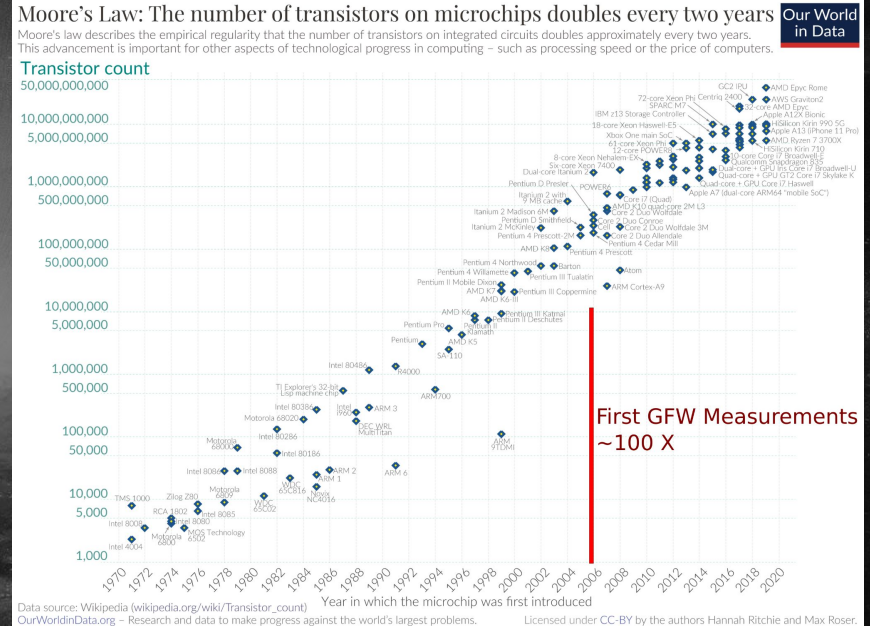
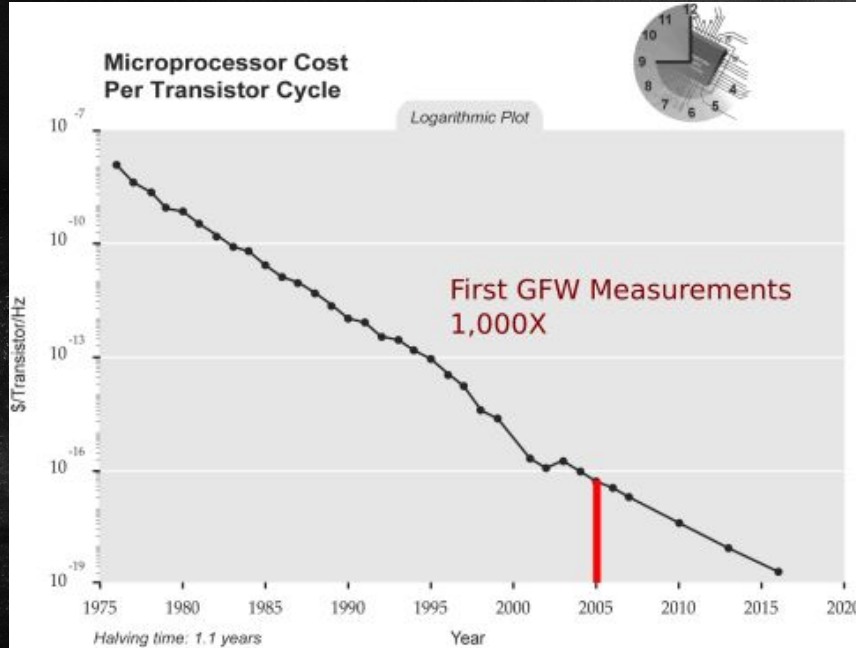


# Why?

- ISP Ecosystem factors:
  - Thousands of ISPs/network operators throughout Russia
  - few in Mainland China
- Economic Factors
  - Transistors in 05-06 were 30000 X more expensive than Today
- Data Scale
  - Number of devices/person increased
  - Amount of data each person generates



# Economics



<https://www.singularity.com/images/charts/MicroProcessCostPerTrans.jpg>, [https://en.wikipedia.org/wiki/Moore%27s\\_law#/media/File:Moore's\\_Law\\_Transistor\\_Count\\_1970-2020.png](https://en.wikipedia.org/wiki/Moore%27s_law#/media/File:Moore's_Law_Transistor_Count_1970-2020.png)

# TSPU In-the-Wild

- 2019: RuNet Law Passed
- 2021: Twitter Blocking
- 2022: Ukraine War
  - Russia begins throttling Google, Twitter, and other news sites.
- Provide details on previous studies
  - How it was done
  - Mapped out TSPU architecture
  - Identified new bypass strategies



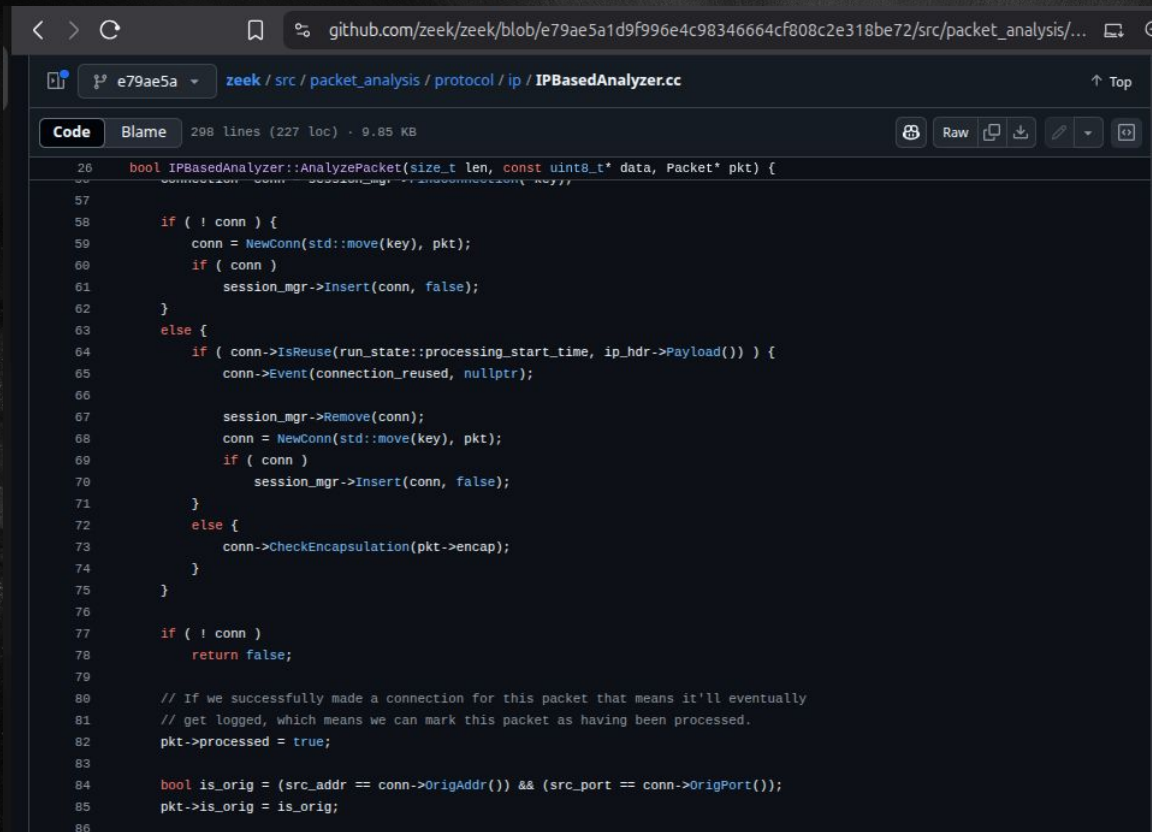
# Connection Tracking

```
< >
github.com/zeeke/zeeke/blob/e79e5a1d9996e4c9b346664f6082e318be72/src/packet_analysis/...
e79e5a1 zeeke / src / packet_analysis / protocol / ip / IPBasedAnalyzer.cc
Code Blame 208 Lines (229 loc) · 9.40 KB
26 bool IPBasedAnalyzer::AnalyzePacket(const IcmpHeader & icmp_header, const Packet & pkt) {
27     // ...
28     if ( ! icmp ) {
29         conn = NewConnFromIcmpHeader(icmp_header, pkt);
30         if ( ! conn ) {
31             session_mgr->insert(conn, false);
32         }
33     } else {
34         if ( conn->isUninitiated() ) {
35             conn->setUninitiatedTime(icmp_header.getTime());
36             session_mgr->insert(conn);
37             conn->setUninitiatedTime(icmp_header.getTime());
38             if ( ! conn ) {
39                 session_mgr->insert(conn, false);
40             }
41         } else {
42             conn->checkUninitiatedTime(icmp_header.getTime());
43         }
44     }
45     if ( ! conn ) {
46         return false;
47     }
48     // If we successfully made a connection for this packet then we'll eventually
49     // get logs, which means we can mark this packet as having been processed.
50     pkt->processed = true;
51     bool is_orig = (src_addr == conn->getSrcAddr()) && (src_port == conn->getSrcPort());
52     pkt->is_orig = is_orig;
53 }
54
```

```
< >
github.com/OSDF/fluxdata/blob/master/src/flow.c
master fluxdata / src / flow.c
Code Blame 240 Lines (222 loc) · 40.9 KB
219 int FlowPacket::Reverse(const Flow & f, const Packet & p) {
220     {
221         const int reverse = (f.flags & FLOW_PKT_REVERSE) > 0;
222         if (p->proto == IPPROTO_TCP || p->proto == IPPROTO_UDP || p->proto == IPPROTO_ICMP) {
223             if (FlowPacket::Reverse(p->hdr, p->payload) < 0) {
224                 /* ... */
225                 return TOCLIENT & reverse;
226             } else {
227                 return TOCLIENT & reverse;
228             }
229         } else {
230             if (FlowPacket::Reverse(p->hdr, p->payload) < 0) {
231                 /* ... */
232                 return TOCLIENT & reverse;
233             } else {
234                 return TOCLIENT & reverse;
235             }
236         }
237     }
238     if (p->proto == IPPROTO_TCP || p->proto == IPPROTO_UDP || p->proto == IPPROTO_ICMP) {
239         if (FlowPacket::Reverse(p->hdr, p->payload) < 0) {
240             /* ... */
241             return TOCLIENT & reverse;
242         } else {
243             return TOCLIENT & reverse;
244         }
245     }
246     /* default to server */
247     return TOCLIENT;
248 }
249
```

```
< >
github.com/torvalds/linux/blob/master/net/netfilter/nf_conntrack_proto_tcp.c
master linux / net / netfilter / nf_conntrack_proto_tcp.c
Code Blame 1645 Lines (1470 loc) · 49 KB
92 enum tcp_bit_set {
203     {
204         /* REPLY */
205         /* sND, sSS, sSR, sES, sFW, sOW, sLA, sTW, sCL, sS2 */
206         /* sym */ { sIV, sS2, sIV, sIV, sIV, sIV, sSS, sIV, sS2 },
207         /*
208          * sND -> sIV      Never reached.
209          * sSS -> sS2      Simultaneous open
210          * sS2 -> sS2      Retransmitted simultaneous SYN
211          * sSR -> sIV      Invalid SYN packets sent by the server
212          * sES -> sIV
213          * sFW -> sIV
214          * sOW -> sIV
215          * sLA -> sIV
216          * sTW -> sSS      Reopened connection, but server may have switched role
217          * sCL -> sIV
218          */
219         /* sND, sSS, sSR, sES, sFW, sOW, sLA, sTW, sCL, sS2 */
220         /* symack */ { sIV, sSR, sIO, sIO, sIO, sIO, sIO, sIO, sSR },
221         /*
222          * sSS -> sSR      Standard open.
223
```

# Connection Tracking: Zeek



The screenshot displays a GitHub web interface for the file `IPBasedAnalyzer.cc` within the `zeek / src / packet_analysis / protocol / ip` directory. The file is 298 lines long, with 227 lines of code and a size of 9.85 KB. The code is written in C++ and implements the `AnalyzePacket` method for the `IPBasedAnalyzer` class. The method takes a packet length, data, and a packet object as input. It checks if a connection already exists for the packet. If it does, it inserts the connection into a session manager. If not, it checks if the connection is a reuse of a previous one. If it is, it removes the old connection and inserts the new one. If not, it checks the encapsulation and inserts the connection. The method returns `false` if no connection is found, and `true` if a connection is found and processed. It also updates the packet's `processed` flag and the `is_orig` flag.

```
26 bool IPBasedAnalyzer::AnalyzePacket(size_t len, const uint8_t* data, Packet* pkt) {
27     // connection = conn = session_mgr->FindConnection(key);
28
29     if ( ! conn ) {
30         conn = NewConn(std::move(key), pkt);
31         if ( conn )
32             session_mgr->Insert(conn, false);
33     }
34     else {
35         if ( conn->IsReuse(run_state::processing_start_time, ip_hdr->Payload()) ) {
36             conn->Event(connection_reused, nullptr);
37
38             session_mgr->Remove(conn);
39             conn = NewConn(std::move(key), pkt);
40             if ( conn )
41                 session_mgr->Insert(conn, false);
42         }
43         else {
44             conn->CheckEncapsulation(pkt->encap);
45         }
46     }
47
48     if ( ! conn )
49         return false;
50
51     // If we successfully made a connection for this packet that means it'll eventually
52     // get logged, which means we can mark this packet as having been processed.
53     pkt->processed = true;
54
55     bool is_orig = (src_addr == conn->OrigAddr()) && (src_port == conn->OrigPort());
56     pkt->is_orig = is_orig;
```



# Connection Tracking: Zeek

```
bool is_orig = (src_addr == conn->OrigAddr()) && (src_port == conn->OrigPort());  
pkt->is_orig = is_orig;
```

```
50     conn = new Conn(std::move(key), pkt);  
51     // ...  
52     session_mgr->Insert(conn, false);  
53  
54     if (conn->IsReuse(run_state->processing, &src_addr, &src_port, &load)) {  
55         conn->SetConn(connection_reused, false);  
56  
57         session_mgr->Remove(conn);  
58         conn = new Conn(std::move(key), pkt);  
59         if (conn) {  
60             session_mgr->Insert(conn, false);  
61         }  
62     }  
63     conn->SetConn(connection_reused, false);  
64     conn->SetConn(connection_reused, false);  
65  
66     return true;  
67 }  
68  
69 // If we successfully made a connection for this packet, then we can  
70 // mark this packet as having been processed.  
71 pkt->processed = true;  
72  
73  
74 bool is_orig = (src_addr == conn->OrigAddr()) && (src_port == conn->OrigPort());  
75 pkt->is_orig = is_orig;
```

# Connection Tracking: Suricata

github.com/OISF/suricata/blob/master/src/flow.c#L279

master ▾ suricata / src / flow.c

[Code](#) [Blame](#) 1460 lines (1252 loc) · 48.9 KB · 

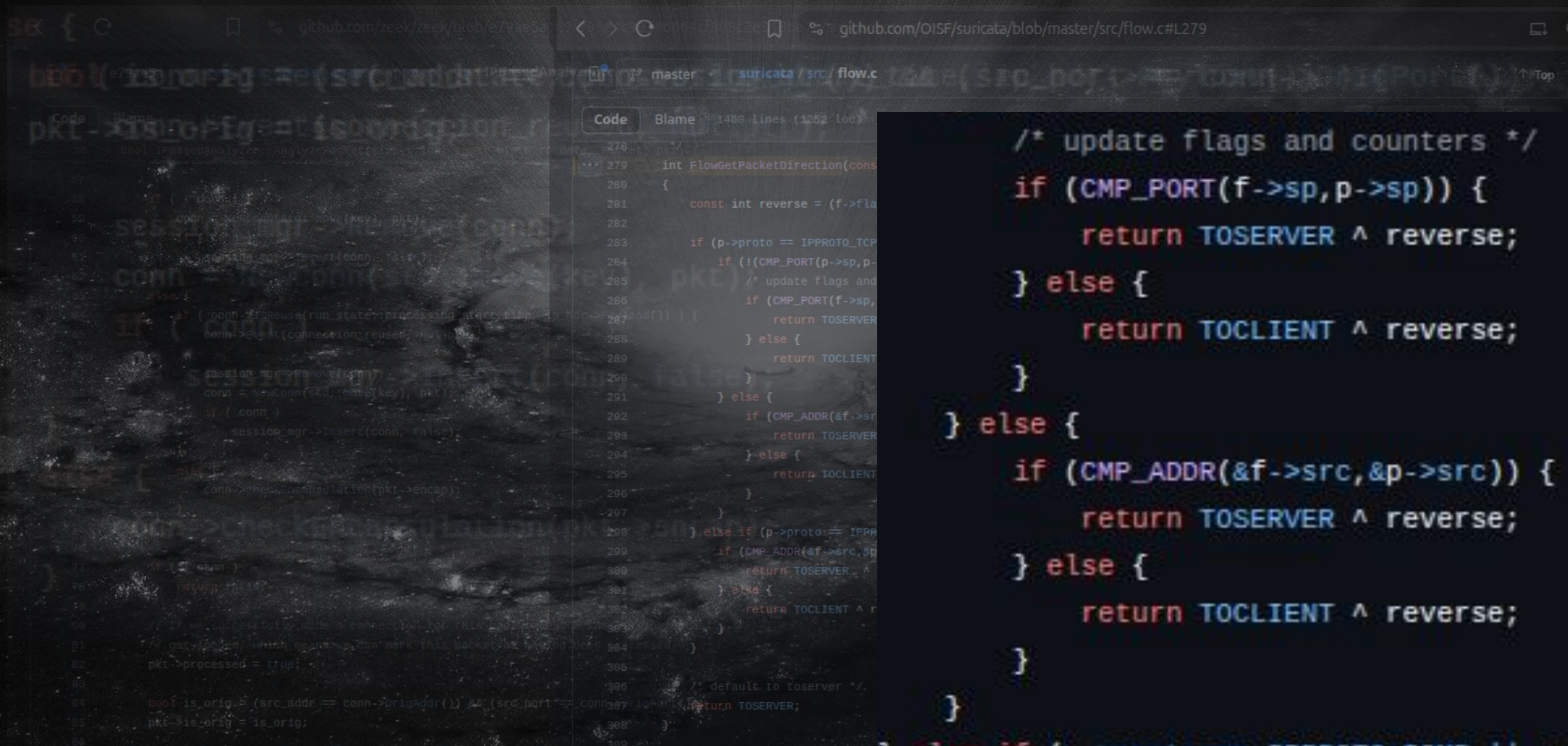
```

278 */
279 int FlowGetPacketDirection(const Flow *f, const Packet *p)
280 {
281     const int reverse = (f->flags & FLOW_DIR_REVERSED) != 0;
282
283     if (p->proto == IPPROTO_TCP || p->proto == IPPROTO_UDP || p->proto == IPPROTO_SCTP) {
284         if (!CMP_PORT(p->sp, p->dp)) {
285             /* update flags and counters */
286             if (CMP_PORT(f->sp, p->sp)) {
287                 return TOSERVER ^ reverse;
288             } else {
289                 return TOCLIENT ^ reverse;
290             }
291         } else {
292             if (CMP_ADDR(&f->src, &p->src)) {
293                 return TOSERVER ^ reverse;
294             } else {
295                 return TOCLIENT ^ reverse;
296             }
297         }
298     } else if (p->proto == IPPROTO_ICMP || p->proto == IPPROTO_ICMPV6) {
299         if (CMP_ADDR(&f->src, &p->src)) {
300             return TOSERVER ^ reverse;
301         } else {
302             return TOCLIENT ^ reverse;
303         }
304     }
305
306     /* default to toserver */
307     return TOSERVER;
308 }
309

```

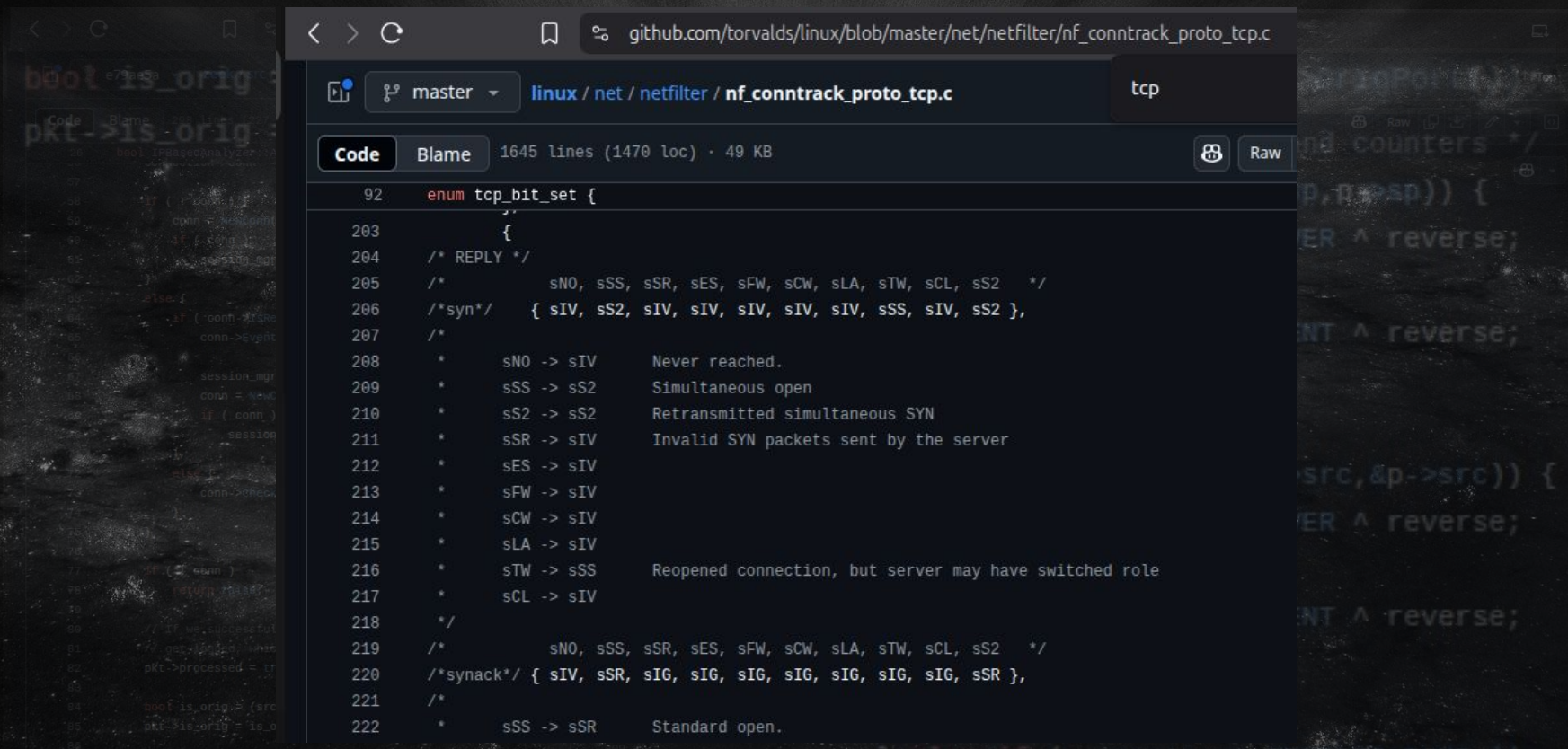


# Connection Tracking: Suricata



```
Code | Blame | 1480 Lines (1252 loc)
278  /*
279  int FlowGetPacketDirection(const
280  {
281      const int reverse = (f->fla
282
283      if (p->proto == IPPROTO_TCP
284          if (!CMP_PORT(p->sp,p
285          /* update flags and
286          if (CMP_PORT(f->sp,
287              return TOSERVER
288          } else {
289              return TOCLIENT
290          }
291      } else {
292          if (CMP_ADDR(&f->sr
293              return TOSERVER
294          } else {
295              return TOCLIENT
296          }
297      }
298      } else if (p->proto == IPP
299          if (CMP_ADDR(&f->src, p
300              return TOSERVER ^
301          } else {
302              return TOCLIENT ^
303          }
304      }
305      /* default to toserver */
306      return TOSERVER;
307  }
```

# Connection Tracking: Netfilter in Linux



The screenshot displays the GitHub web interface for the file `nf_conntrack_proto_tcp.c` in the `linux/net/netfilter` directory. The file is 1645 lines long (1470 loc) and 49 KB in size. The `Code` tab is selected, showing the following code snippet:

```
92  enum tcp_bit_set {
203      {
204      /* REPLY */
205      /*      sNO, sSS, sSR, sES, sFW, sCW, sLA, sTW, sCL, sS2 */
206      /*syn*/  { sIV, sS2, sIV, sIV, sIV, sIV, sSS, sIV, sS2 },
207      /*
208      *      sNO -> sIV      Never reached.
209      *      sSS -> sS2      Simultaneous open
210      *      sS2 -> sS2      Retransmitted simultaneous SYN
211      *      sSR -> sIV      Invalid SYN packets sent by the server
212      *      sES -> sIV
213      *      sFW -> sIV
214      *      sCW -> sIV
215      *      sLA -> sIV
216      *      sTW -> sSS      Reopened connection, but server may have switched role
217      *      sCL -> sIV
218      */
219      /*      sNO, sSS, sSR, sES, sFW, sCW, sLA, sTW, sCL, sS2 */
220      /*synack*/ { sIV, sSR, sIG, sIG, sIG, sIG, sIG, sIG, sSR },
221      /*
222      *      sSS -> sSR      Standard open.
```

The background of the slide features a dark, textured image with faint, glowing lines and text, suggesting a network or data flow theme.



# Connection Tracking: Netfilter in Linux

Reply

Simultaneous  
Open

/\* REPLY \*/

/\* sNO, sSS, sSR, sES, sFW, sCW, sLA, sTW, sC

/\*syn\*/ { sIV, sS2, sIV, sIV, sIV, sIV, sIV, sSS, sI

/\*

\* sNO > sIV Never reached.

\* sSS -> sS2 Simultaneous open

\* sS2 > sS2 Retransmitted simultaneous SYN

# Connection Tracking: Direction

```
bool is_orig = (src_addr == conn->OrigAddr()) && (src_port == conn->OrigPort());
pkt->is_orig = is_orig;
```

```

50      conn = NewConn(std::move(key), pkt);
51      if (conn) {
52          session_mgr->insert(conn, false); // REPLY
53      }
54      else {
55          if (conn->isReuse(run_state::processing_start_time, ip_hdr->src_addr)) {
56              conn->CountConnectionPoliciesAndToServer();

```

```
/* REPLY */
```

```

/*          sNO, sSS, sSR, sES, sFW, sCI
/*syn*/      { sIV, sS2, sIV, sIV, sIV, sIV
/*

```

```

*          sNO -> sIV      Never reached.
*          sSS -> sS2      Simultaneous open
*          sS2 -> sS2      Retransmitted sir

```

```
/* update flags and counters */
```

```
if (CMP_PORT(f->sp, p->sp)) {
```

```
    return TOSERVER ^ reverse;
```

```
} else {
```

```
    return TOCLIENT ^ reverse;
```

```
}
```

```
} else {
```

```
    if (CMP_ADDR(&f->src, &p->src)) {
```

```
        return TOSERVER ^ reverse;
```

```
    } else {
```

```
        return TOCLIENT ^ reverse;
```

```
}
```

```
}
```





TSPU



Web Server



Connection Tracking Table

| Orig | Reply | TCP State |
|------|-------|-----------|
|------|-------|-----------|



**Connection State Table  
(for TSPU)**



TSPU



Web Server



**“Client”**

Connection Tracking Table

| Orig | Reply | TCP State |
|------|-------|-----------|
|------|-------|-----------|





TSPU



Web Server



**“Server”**



Connection Tracking Table

| Orig | Reply | TCP State |
|------|-------|-----------|
|------|-------|-----------|



TSPU



Web Server



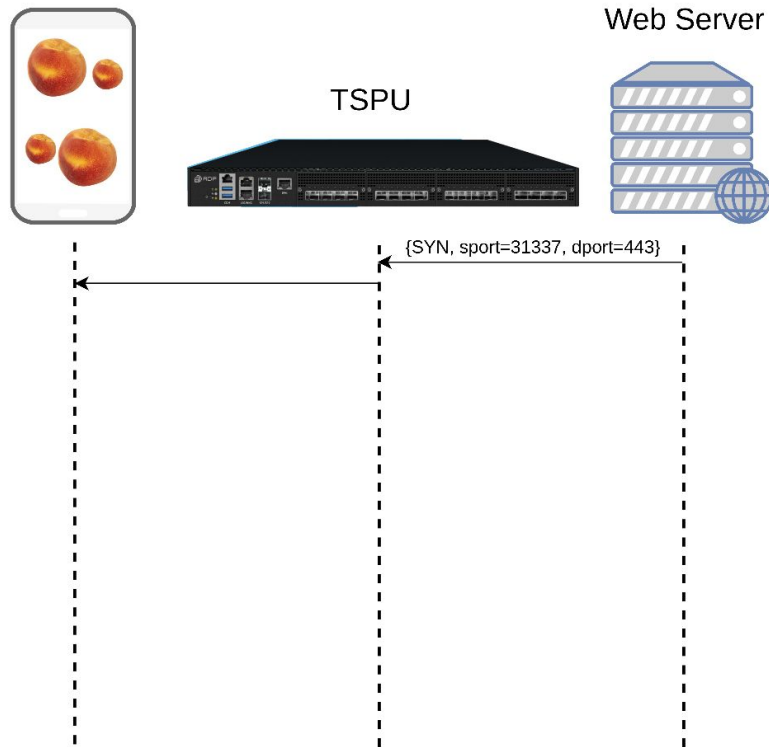
**TCP State**



Connection Tracking Table

| Orig | Reply | TCP State |
|------|-------|-----------|
|------|-------|-----------|





Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |

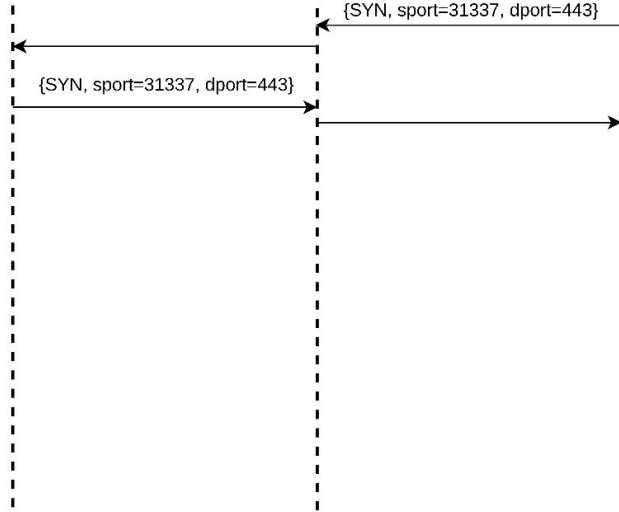
**SYN\_SENT**



TSPU



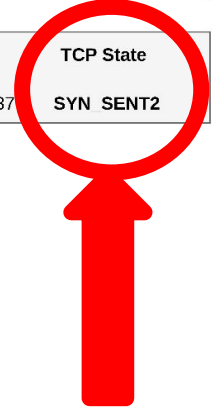
Web Server



### Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |



**SYN\_SENT2**

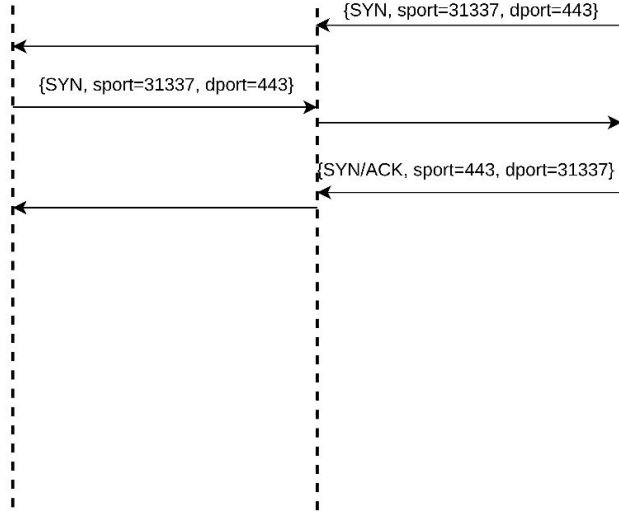




TSPU



Web Server

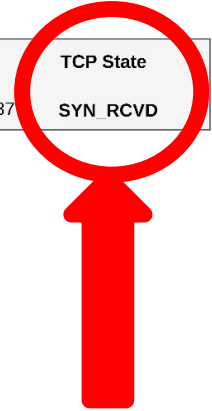


Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |



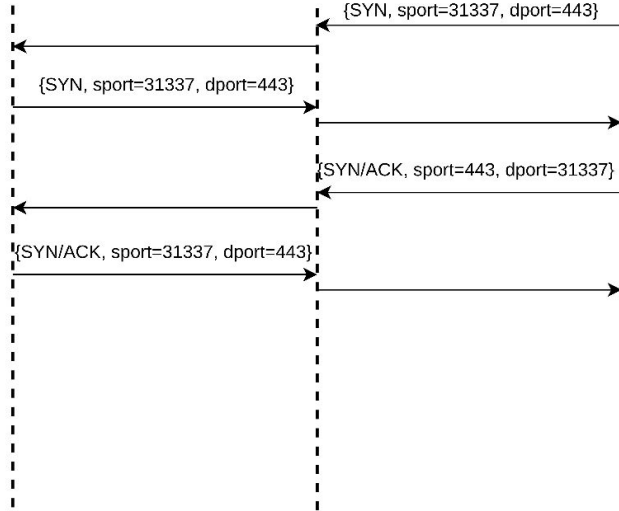
**SYN\_RCVD**



TSPU



Web Server



Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

SYN\_RCVD

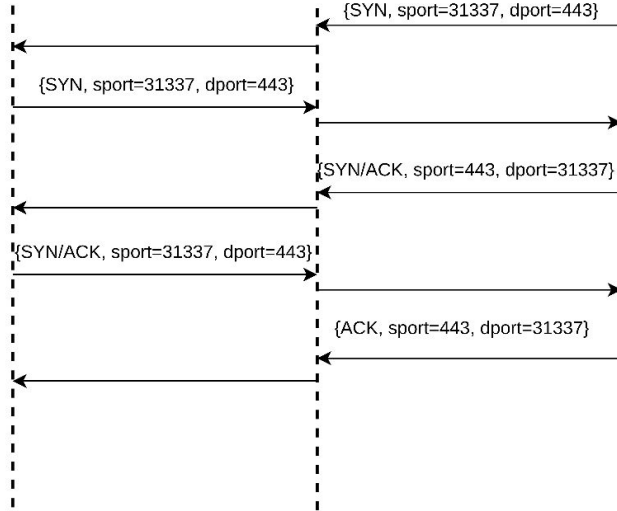




TSPU



Web Server



Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |

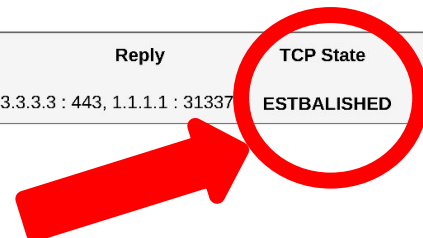
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

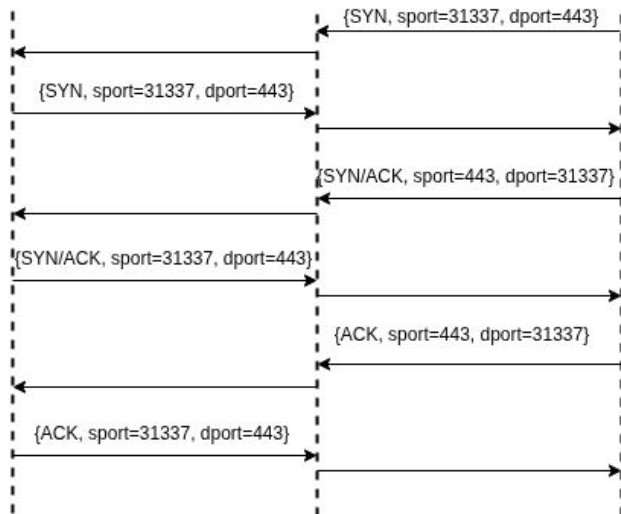
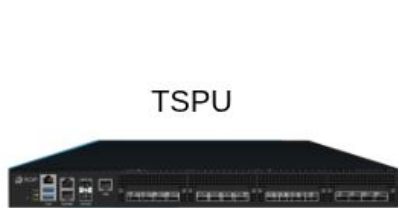
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |

ESTABLISHED





## Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |

*Server-side*

*Bypass:*

Simultaneous Open



Internal Client



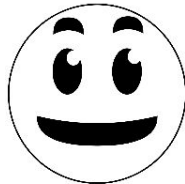
TSPU



Web server



External Client



Connection Tracking Table

| Orig | Reply | TCP State |
|------|-------|-----------|
|      |       | None      |

Internal Client



TSPU



Web server



External Client



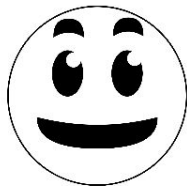
{SYN, sport=31337, dport=443}

Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |

SYN\_SENT

Internal Client



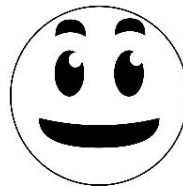
TSPU



Web server



External Client



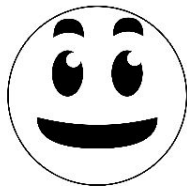
Connection Tracking Table

| Orig                               | Reply                               | TCP State |
|------------------------------------|-------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443 , 1.1.1.1 : 31337 } | SYN_SENT  |
| Orig                               | Reply                               | TCP State |
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 }  | SYN_SENT2 |

SYN\_SENT2



Internal Client



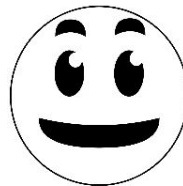
TSPU



Web server



External Client



### Connection Tracking Table

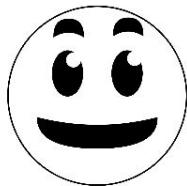
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

**SYN\_RCVD**

Internal Client



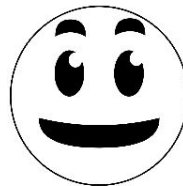
TSPU



Web server



External Client



### Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

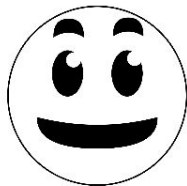
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

**SYN\_RCVD**



Internal Client



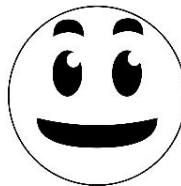
TSPU



Web server



External Client



### Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

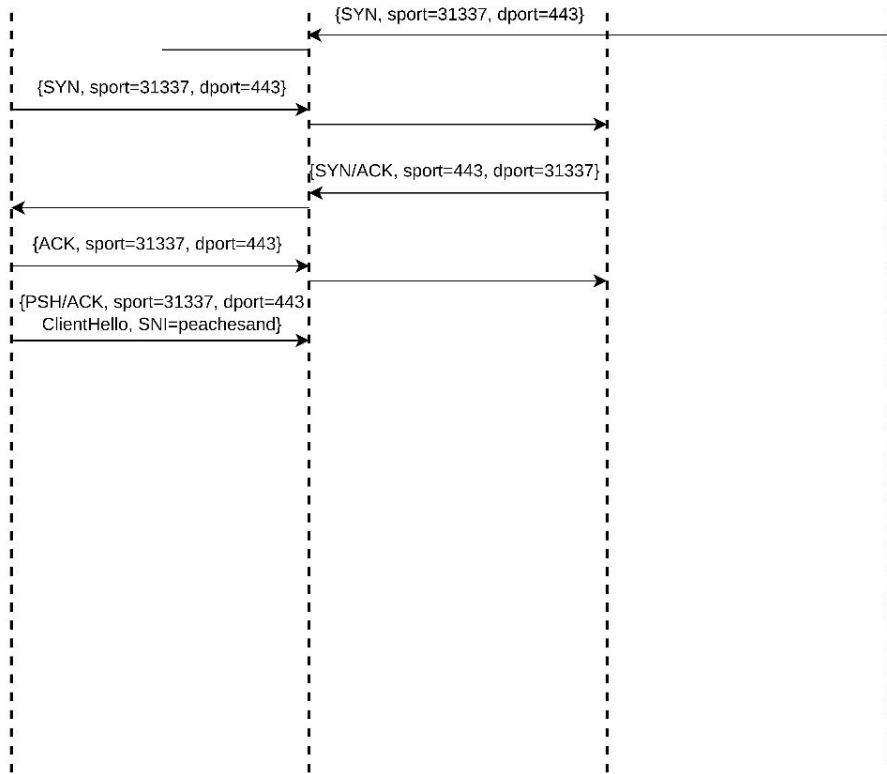
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

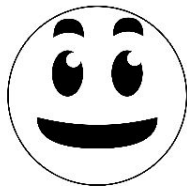
| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |

**ESTABLISHED**





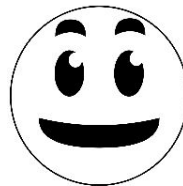
Internal Client



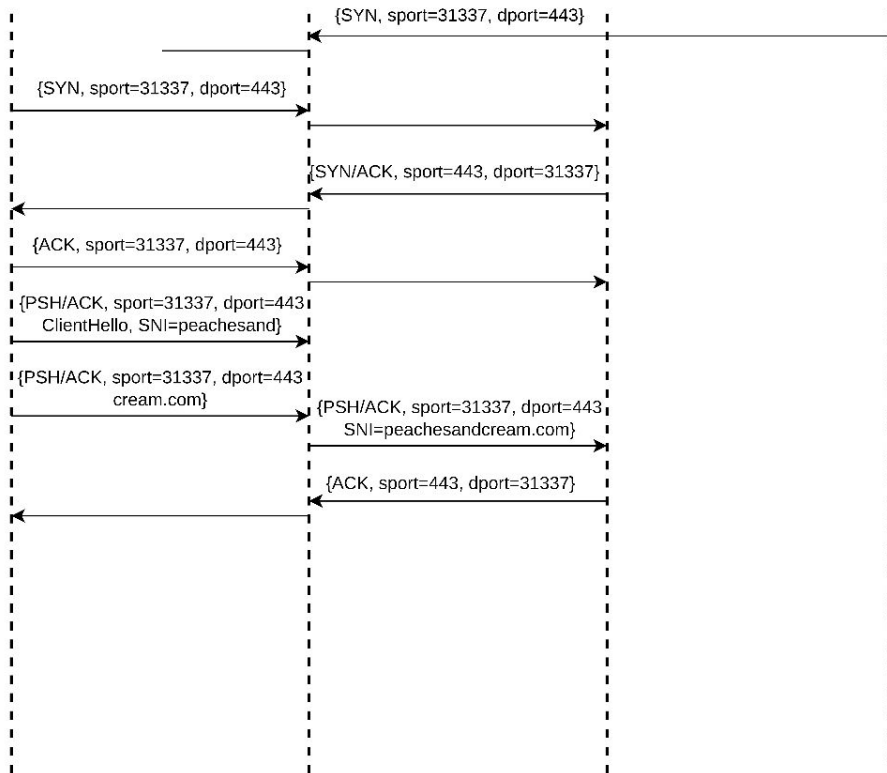
Web server



External Client



TSPU



## Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |

Internal Client



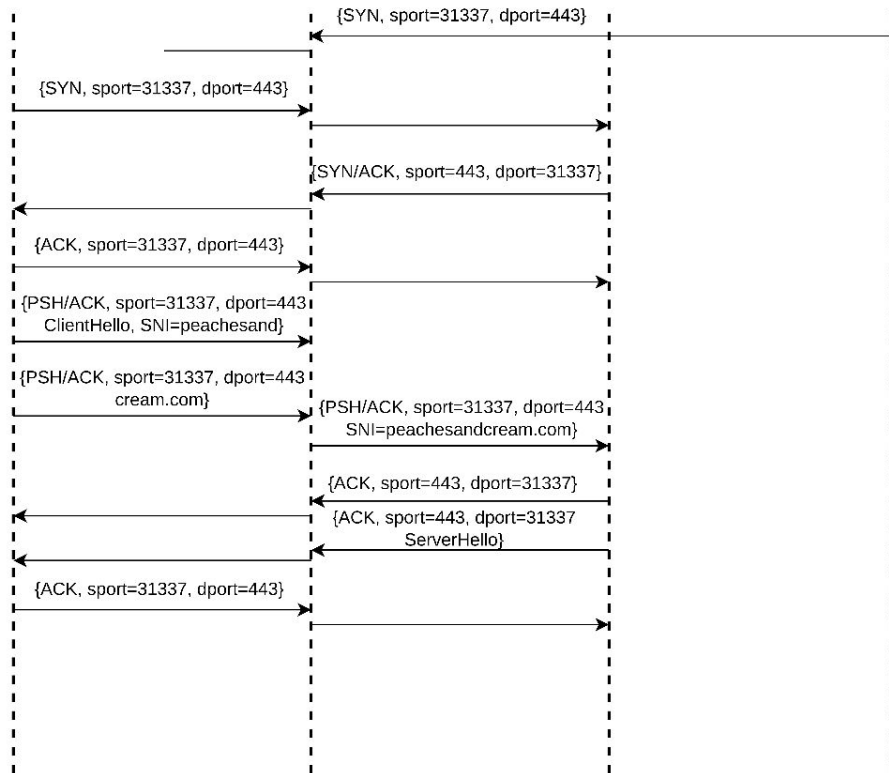
Web server



External Client



TSPU



## Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |

Internal Client



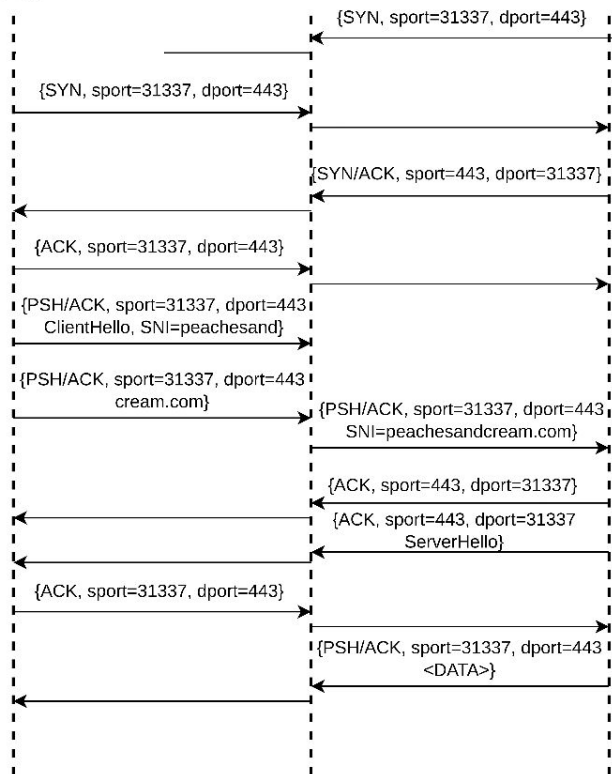
TSPU



Web server



External Client



### Connection Tracking Table

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT  |

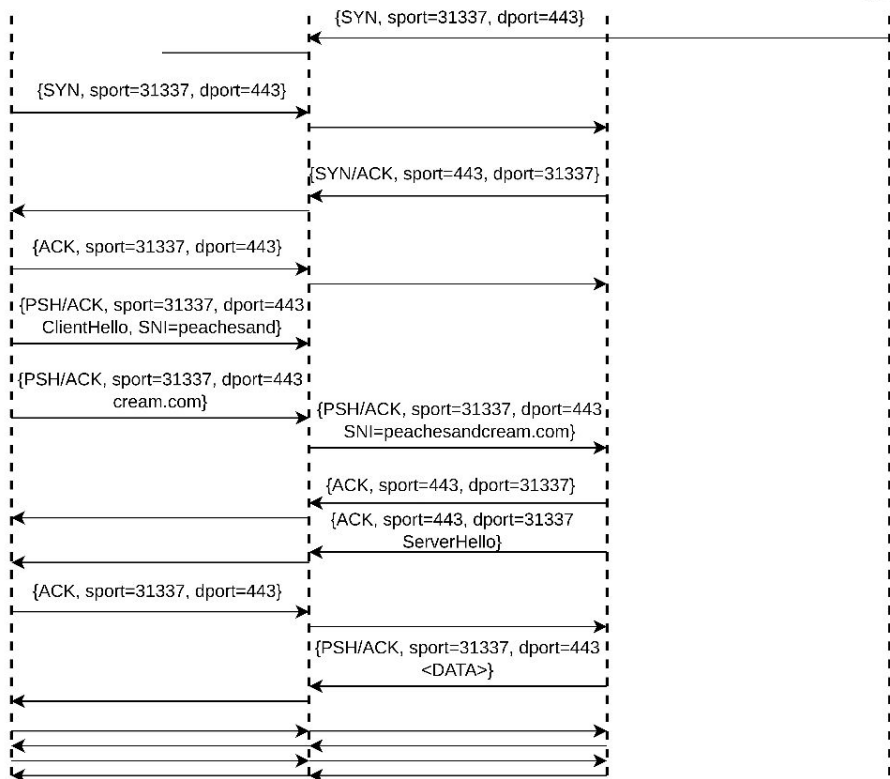
| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State   |
|------------------------------------|------------------------------------|-------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | ESTBALISHED |





## Connection Tracking Table

| Orig                              | Reply                             | TCP State |
|-----------------------------------|-----------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443} | { 3.3.3.3 : 443, 1.1.1.1 : 31337} | SYN_SENT  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_SENT2 |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State |
|------------------------------------|------------------------------------|-----------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | SYN_RCVD  |

| Orig                               | Reply                              | TCP State          |
|------------------------------------|------------------------------------|--------------------|
| { 1.1.1.1 : 31337, 3.3.3.3 : 443 } | { 3.3.3.3 : 443, 1.1.1.1 : 31337 } | <b>ESTABLISHED</b> |

# Other Implications: VPN Security



# Other Implications: VPN Security

breakpointingbad.com/2021/09/08/Port-Shadows-via-Network-Alchemy.html

petsymposium.org/popets/2024/popets-2024-0070.pdf

Tracking Frameworks as used by Virtual Pri... 1 / 18 100%

## Attacking Connection Tracking Frameworks as used by Virtual Private Networks

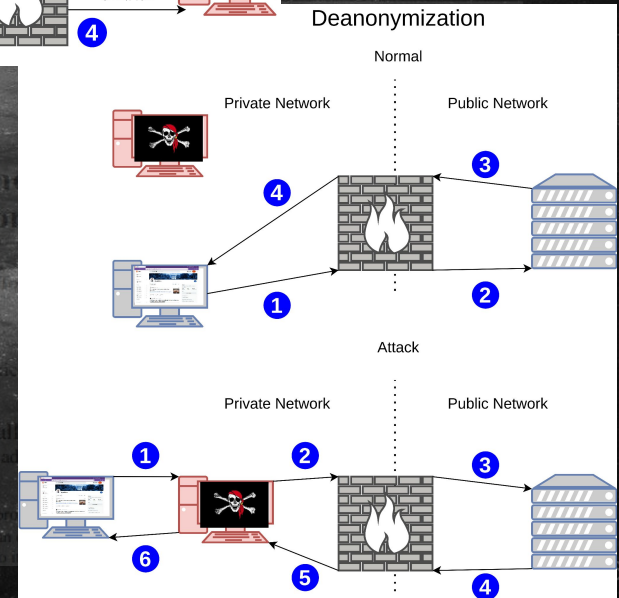
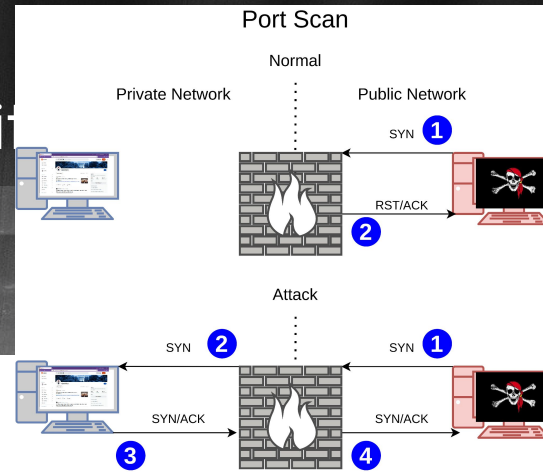
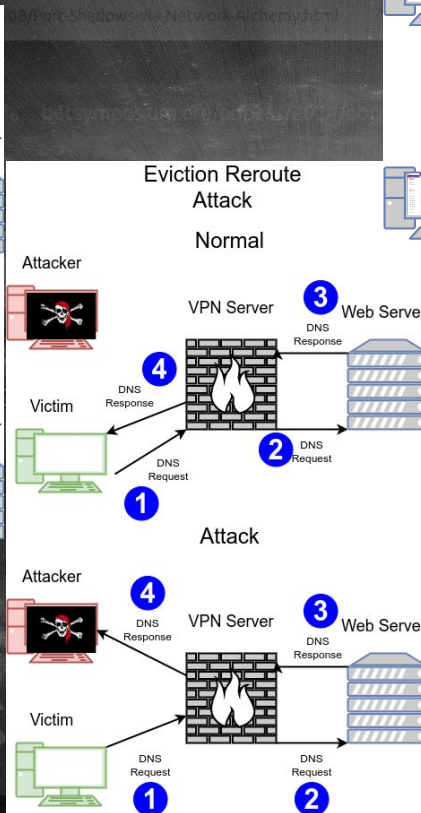
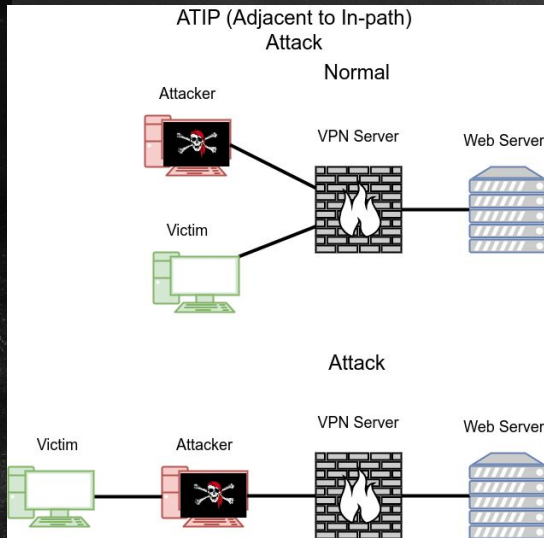
|                                                                  |                                                       |                                       |
|------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------|
| Benjamin Mixon-Baca<br>ASU/Breakpointing Bad<br>bmixonba@asu.edu | Jeffrey Knockel<br>Citizen Lab, University of Toronto | Diwen Xue<br>University of Michigan   |
| Tarun Ayyagari<br>Arizona State University                       | Deepak Kapur<br>University of New Mexico              | Roya Ensafi<br>University of Michigan |
| Jedidiah R. Crandall<br>ASU/Breakpointing Bad                    |                                                       |                                       |

**ABSTRACT**

VPNs (Virtual Private Networks) have become an essential privacy-enhancing technology, particularly for at-risk users such as dissidents to compromise their security. VPN clients encapsulate users' traffic within an encrypted tunnel to the VPN server, which then relays the traffic to its final destination, thus protecting users from potentially



# Other Implications: VPN Security



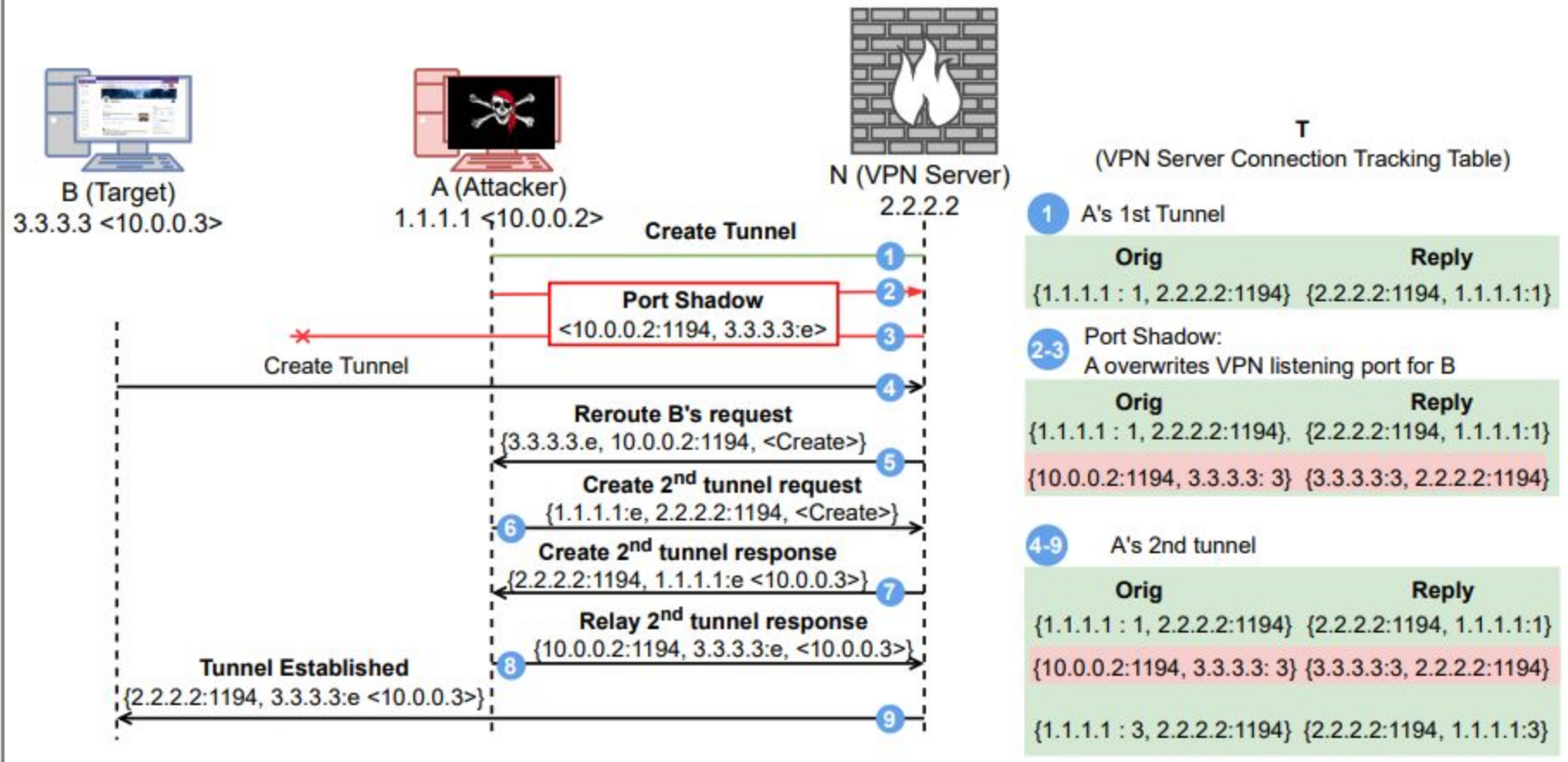


Figure 2: Adjacent-to-in-path attack.



# Other Implications: VPN Security

[petsymposium.org/popets/2024/popets-2024-0070.php](https://petsymposium.org/popets/2024/popets-2024-0070.php)

**(CVE-2021-3773)**

[breakpointingbad.com/2021/09/08/Port-Shadows-via-Network-Alchemy.html](https://breakpointingbad.com/2021/09/08/Port-Shadows-via-Network-Alchemy.html)

**Abstract:** VPNs (Virtual Private Networks) have become an essential privacy-enhancing technology, particularly for at-risk users like dissidents, journalists, NGOs, and others vulnerable to targeted threats. While previous research investigating VPN security has focused on cryptographic strength or traffic leakages, there remains a gap in understanding how lower-level primitives fundamental to VPN operation, like connection tracking, might undermine the security and privacy that VPNs are intended to provide.

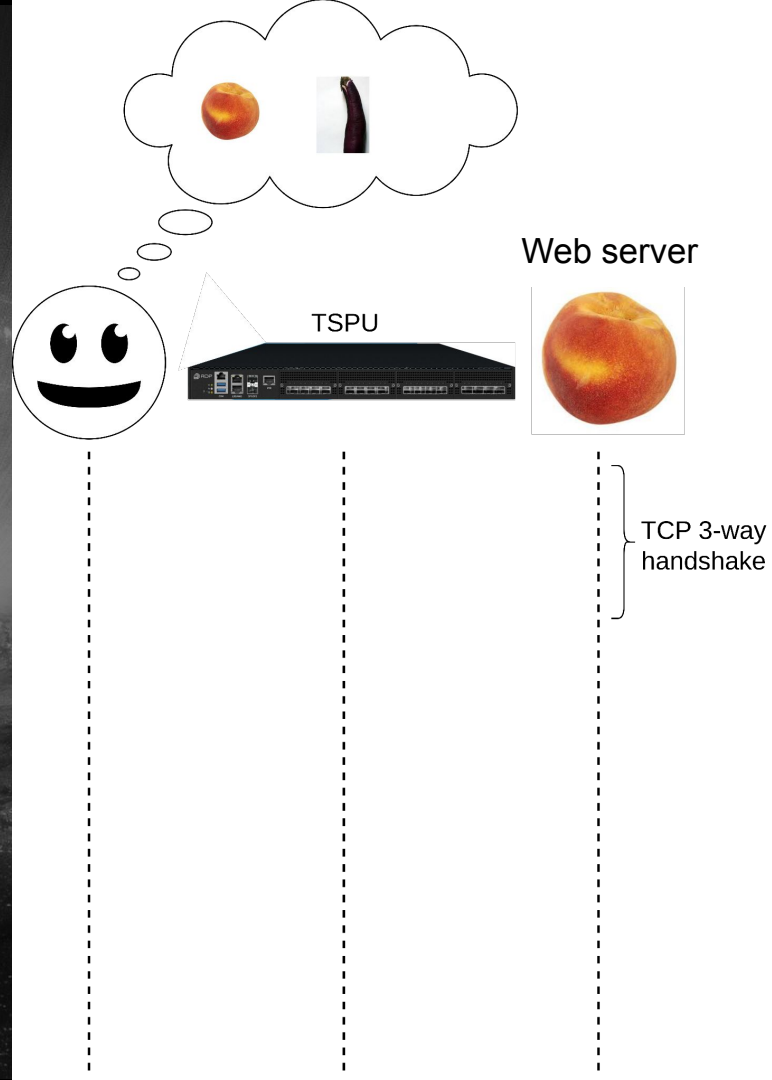
In this paper, we leverage the connection tracking mechanisms used in common operating systems, identifying a novel exploit



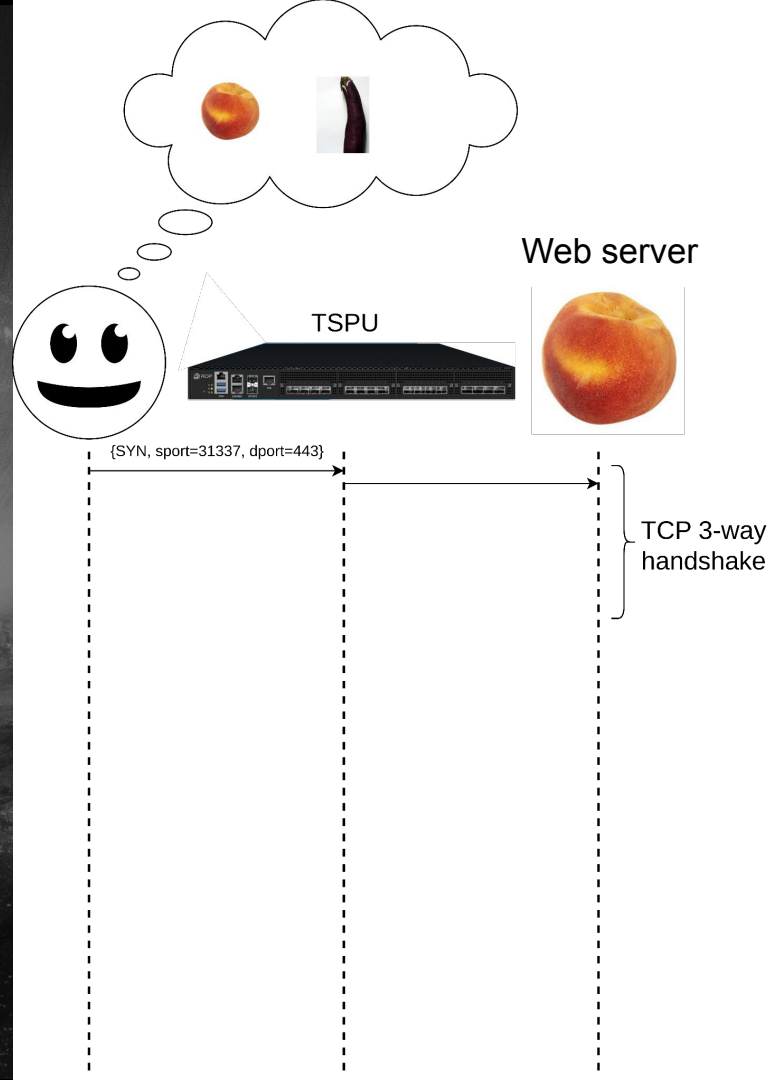
***Client-side Bypass:***

5 ACKS

# TLS- 4 way Handshake

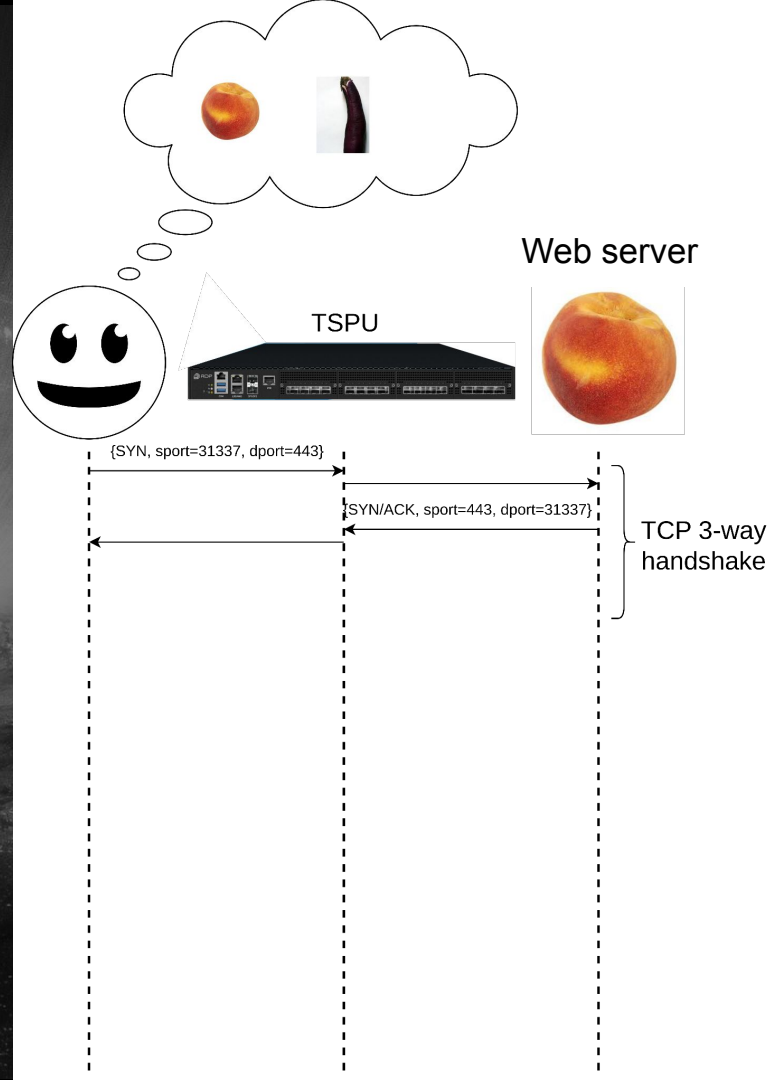


# TLS- 4 way Handshake

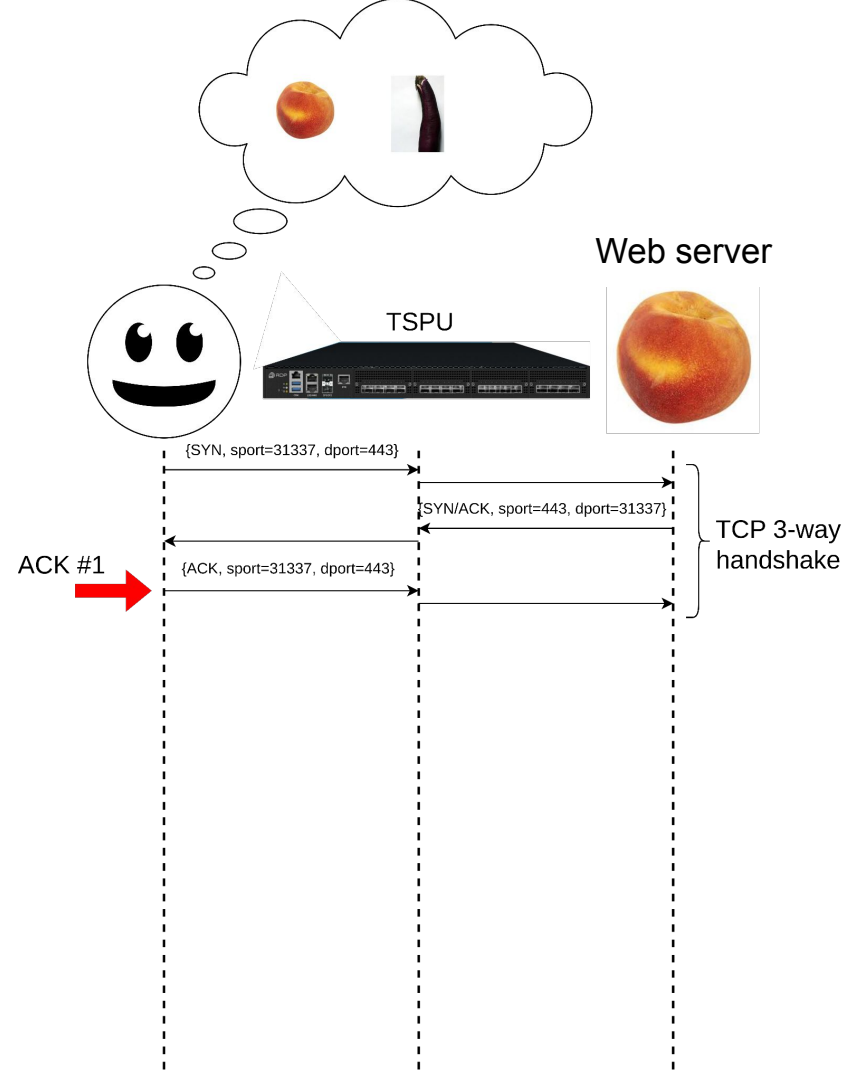




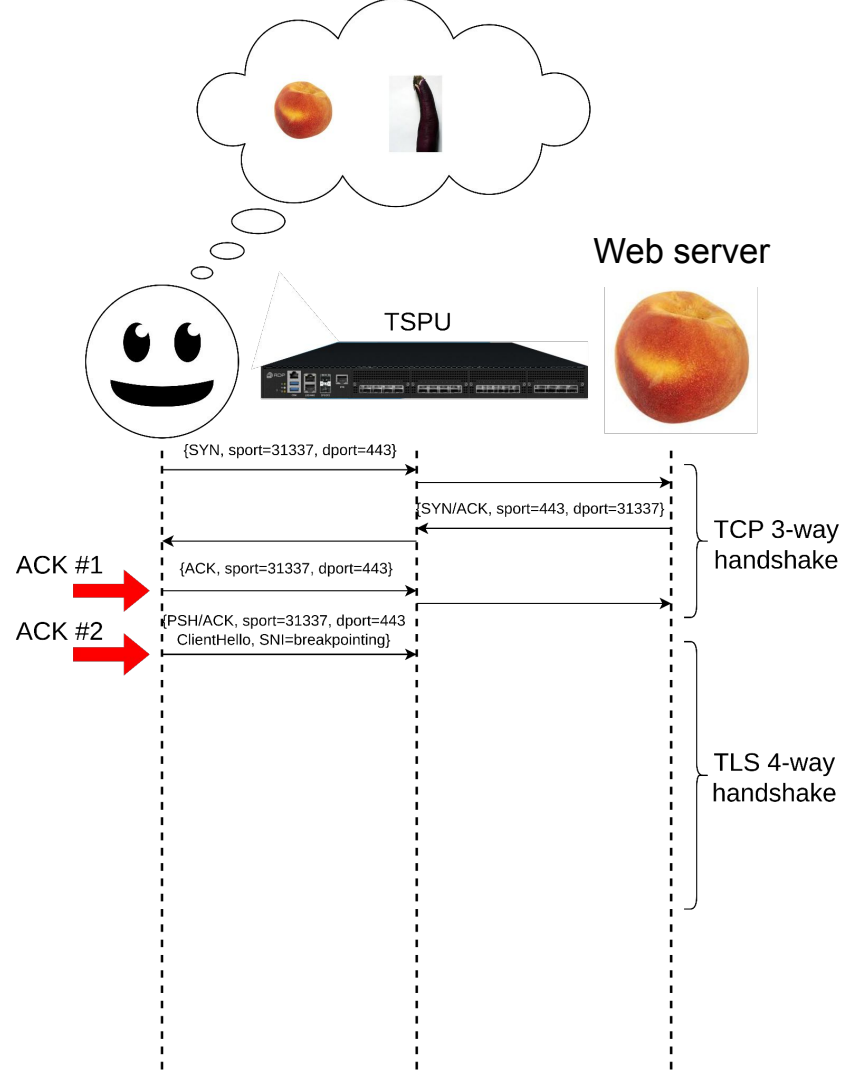
# TLS- 4 way Handshake



# TLS- 4 way Handshake

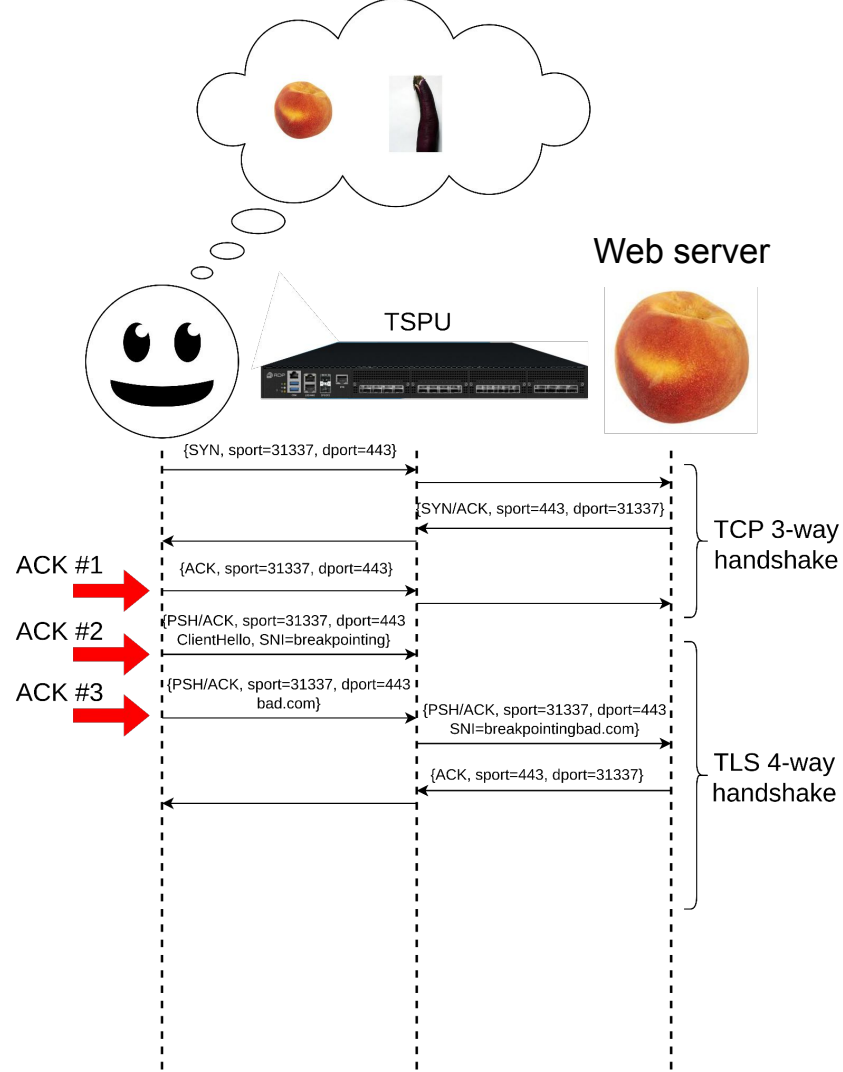


# TLS- 4 way Handshake

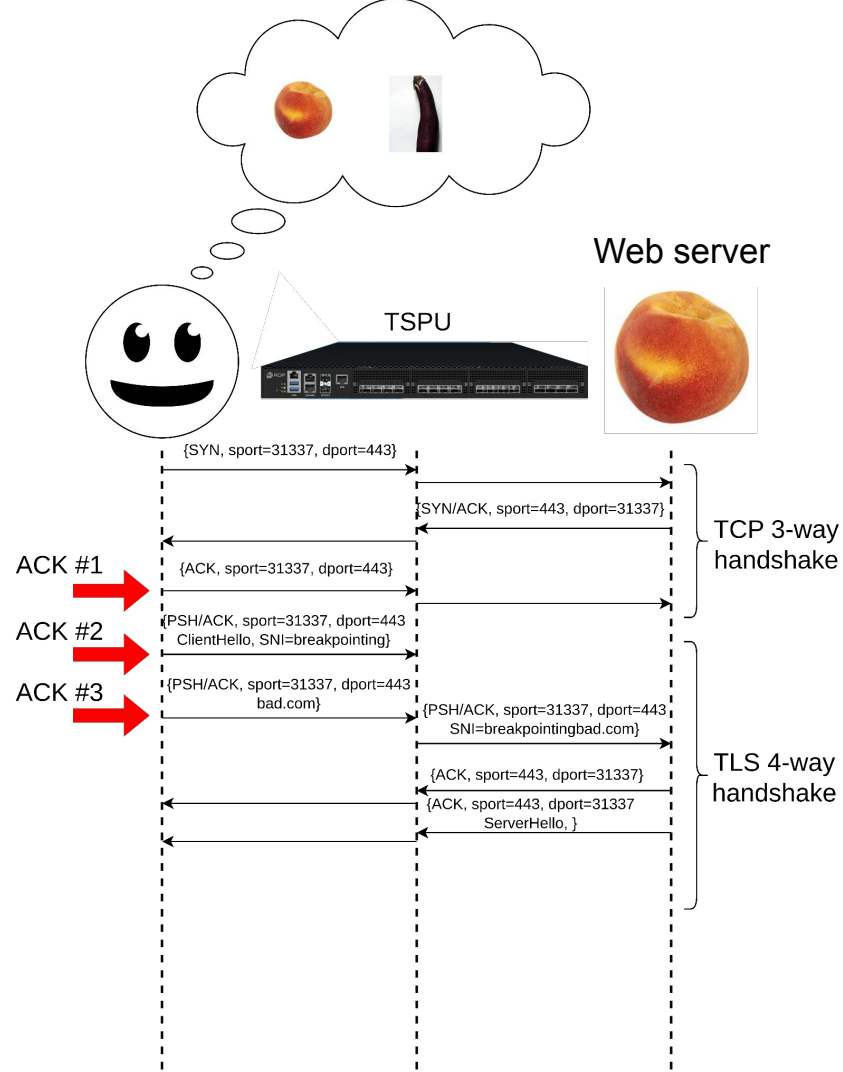




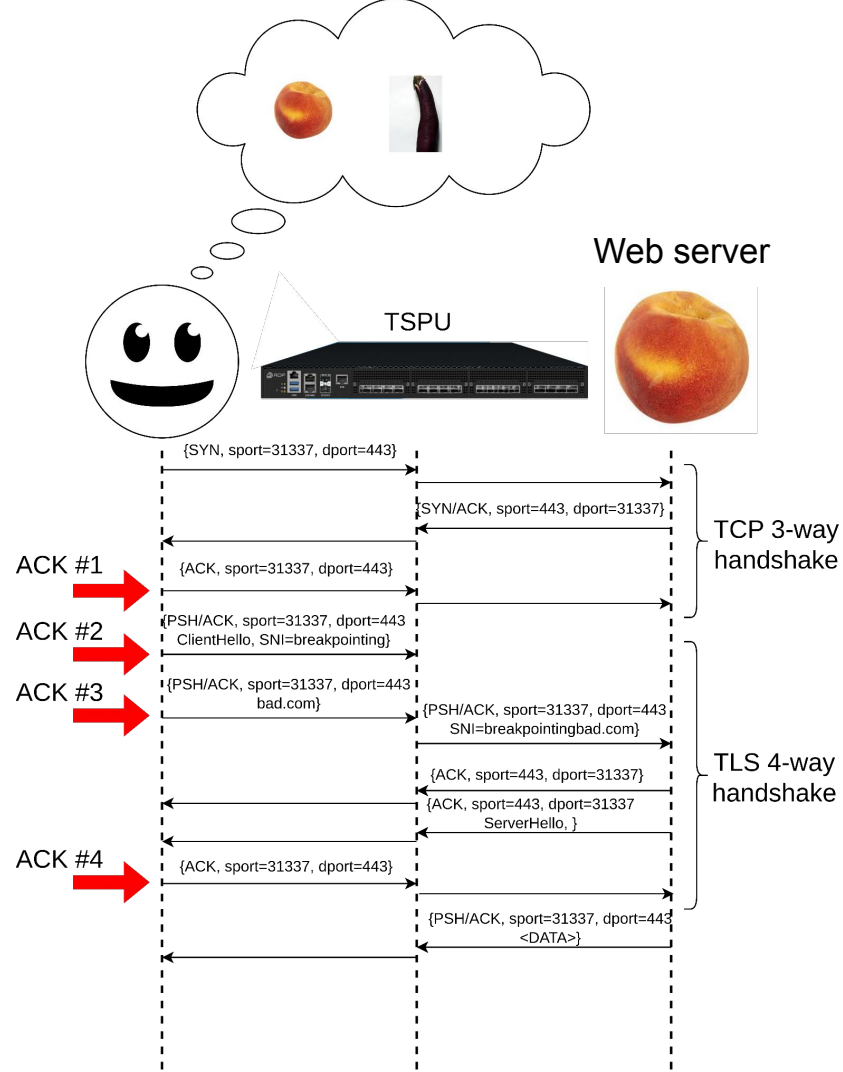
# TLS- 4 way Handshake



# TLS- 4 way Handshake

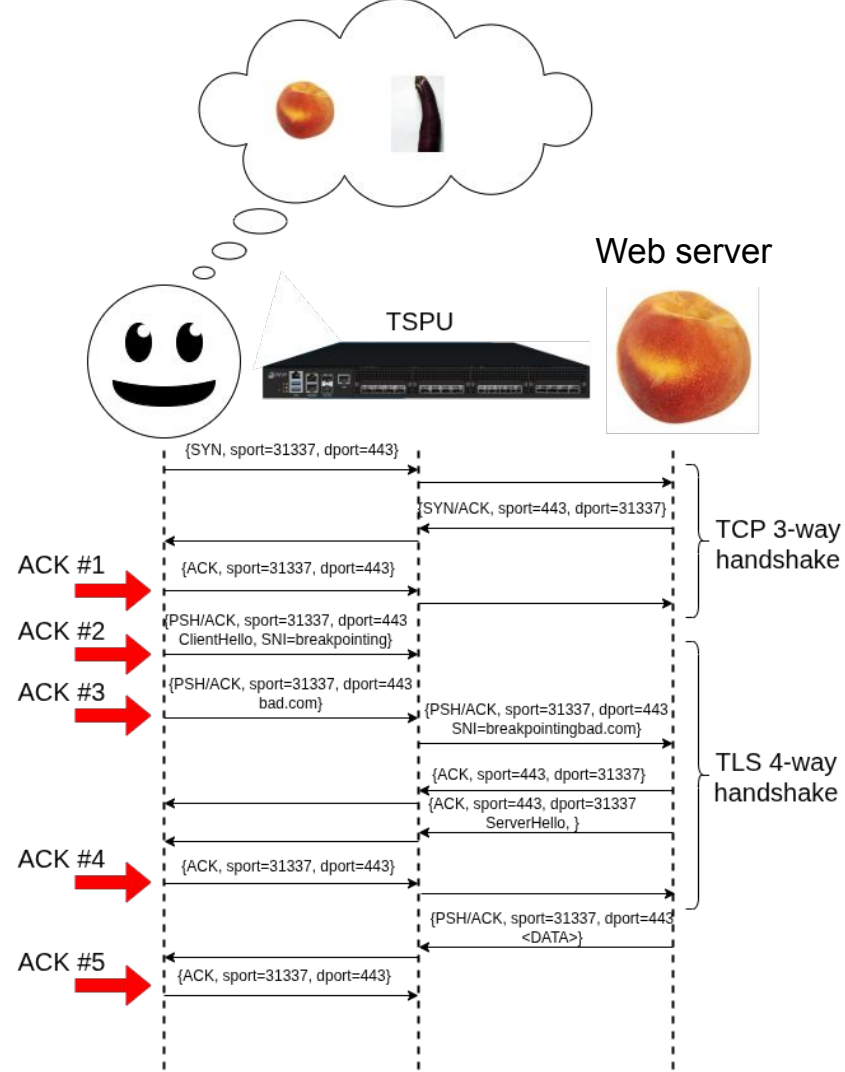


# TLS- 4 way Handshake

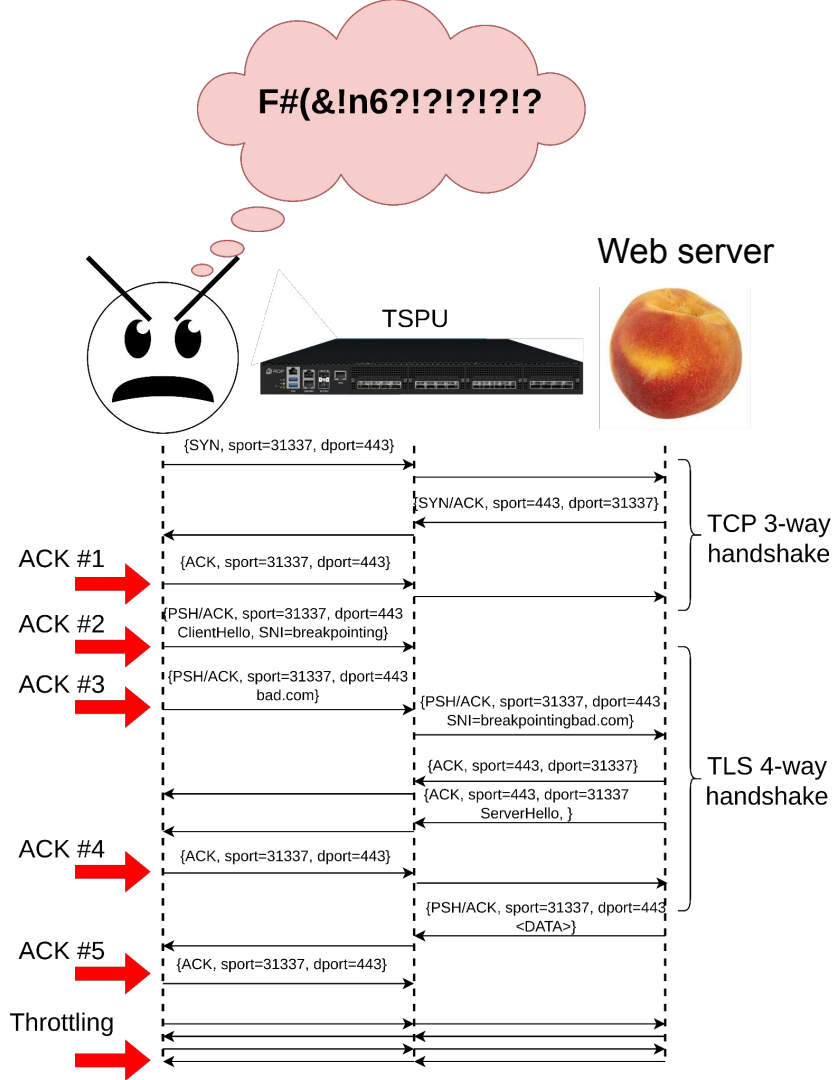




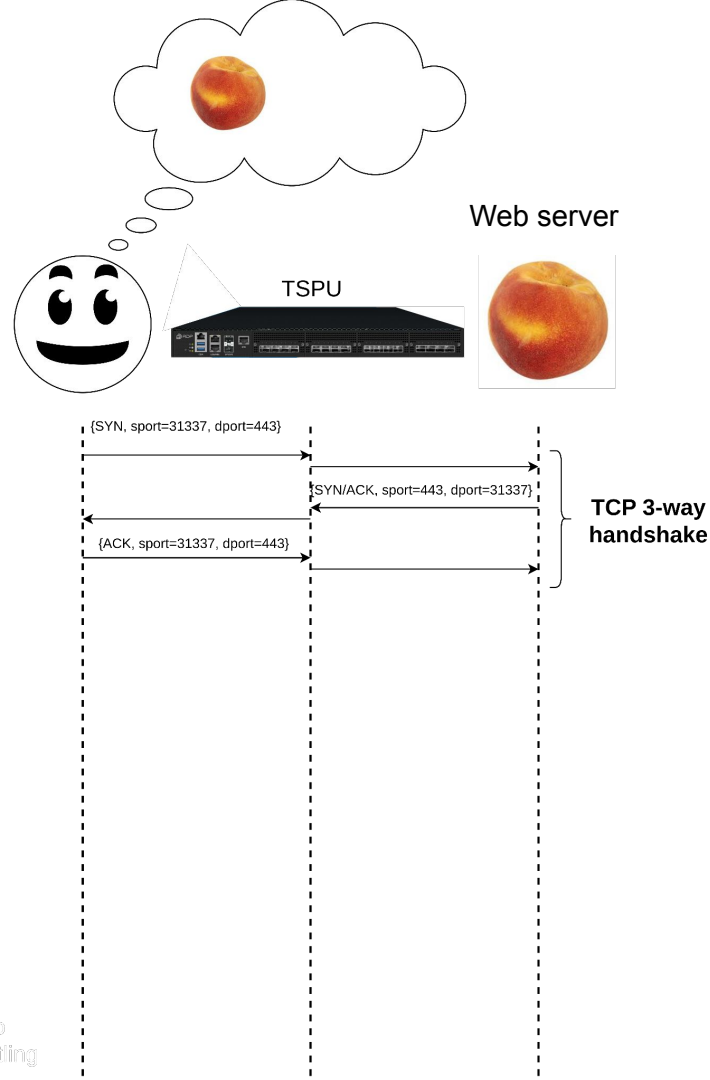
# TLS- 4 way Handshake



# TLS- 4 way Handshake

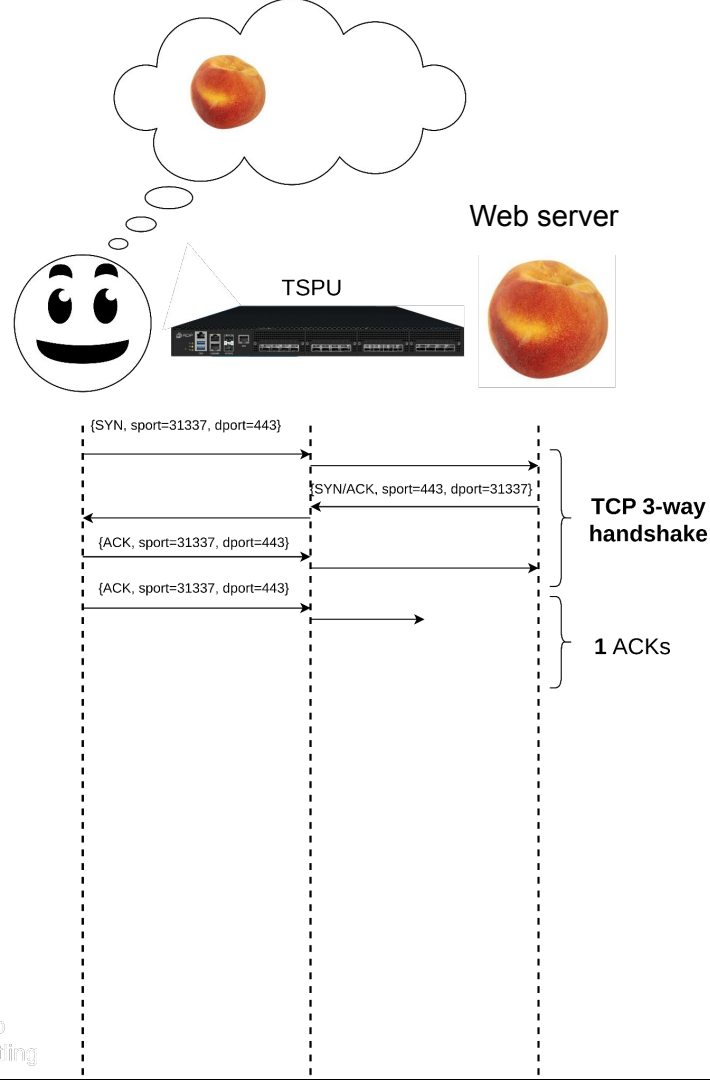


# Client-side Bypass: 5 ACKs

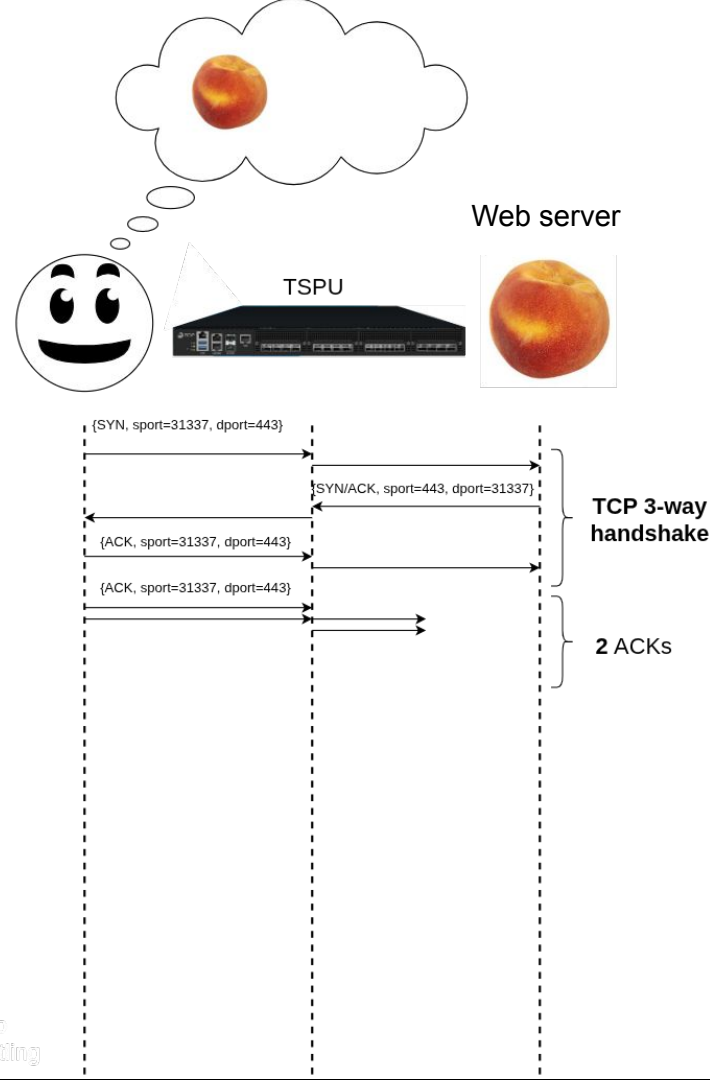




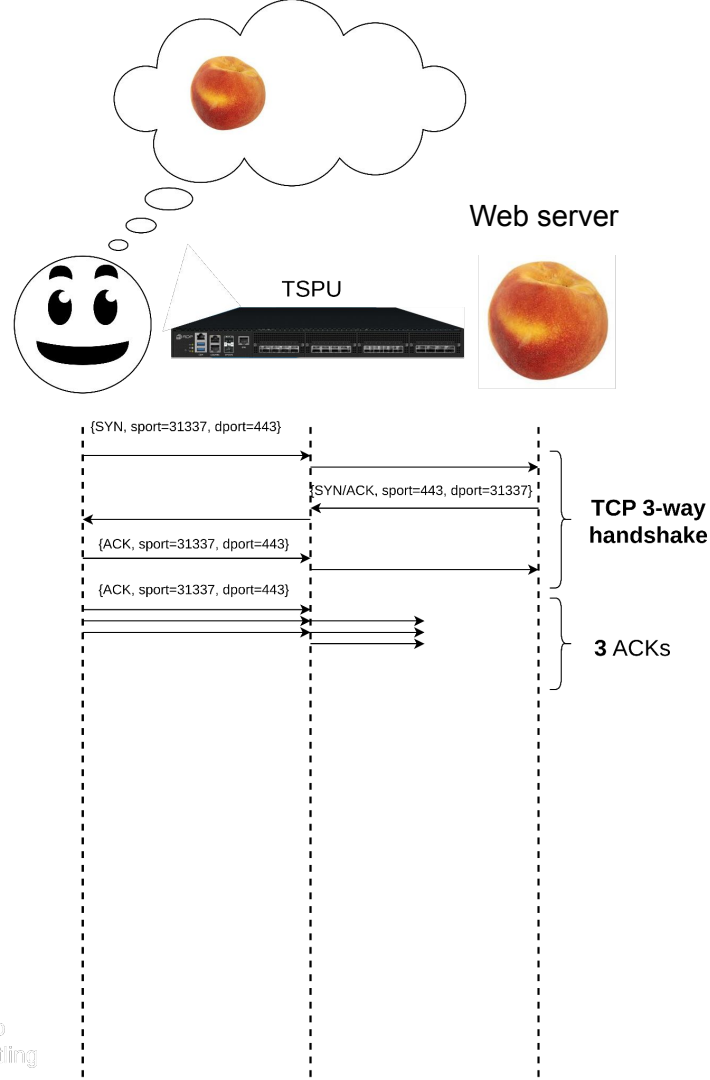
# Client-side Bypass: 5 ACKs



# Client-side Bypass: 5 ACKs

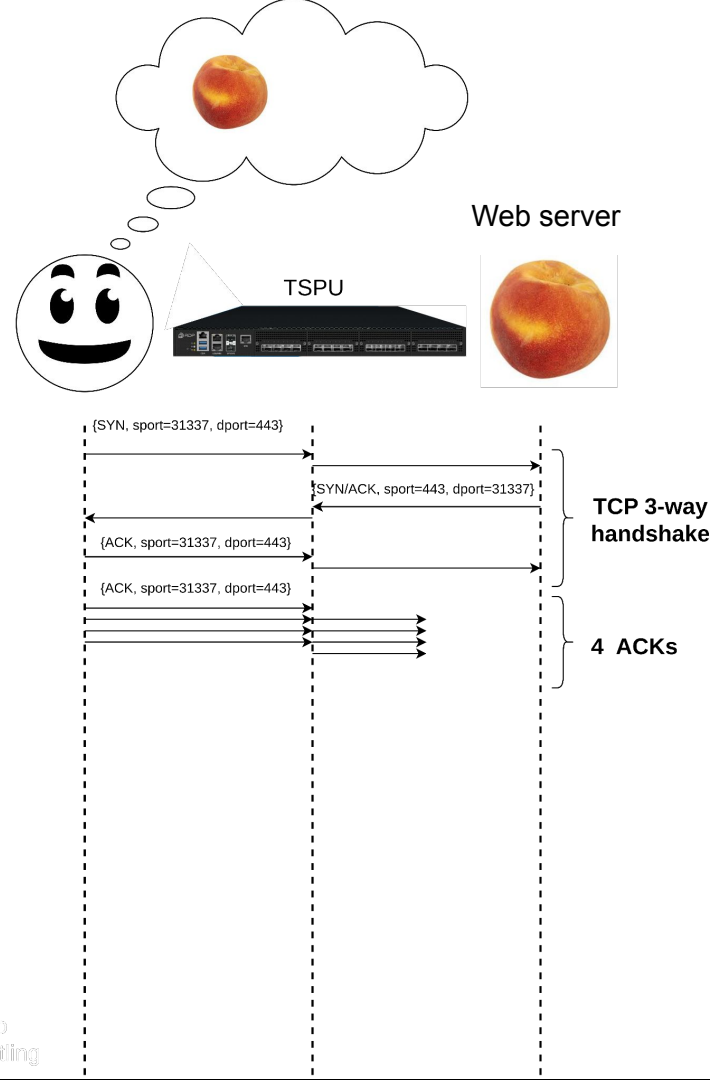


# Client-side Bypass: 5 ACKs

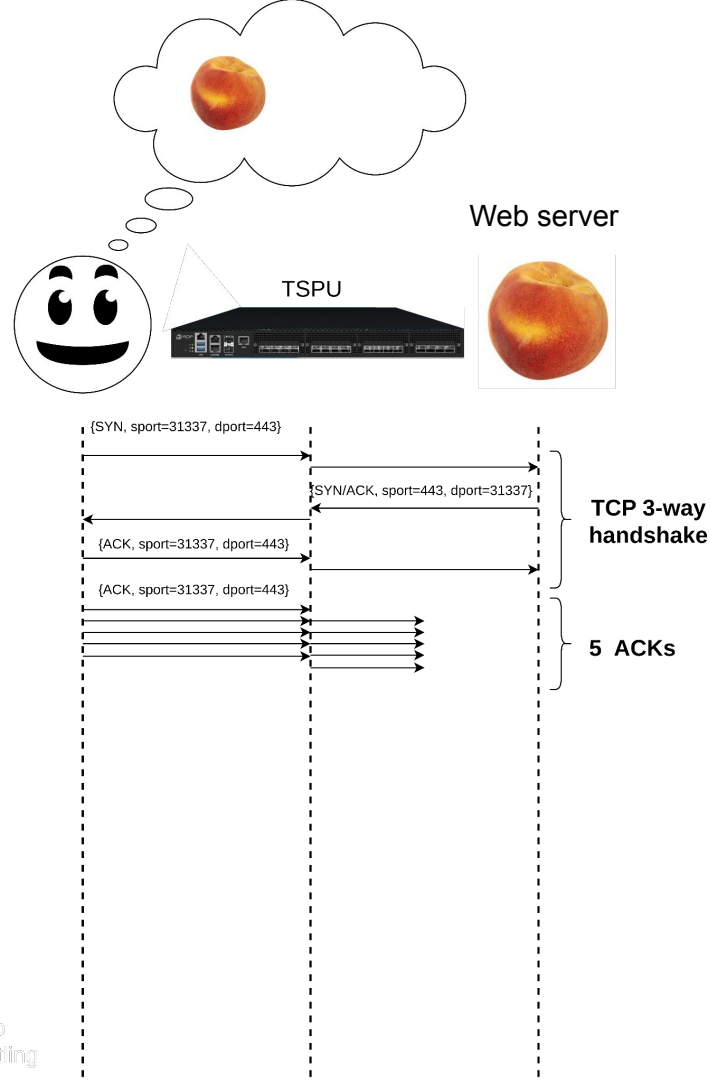




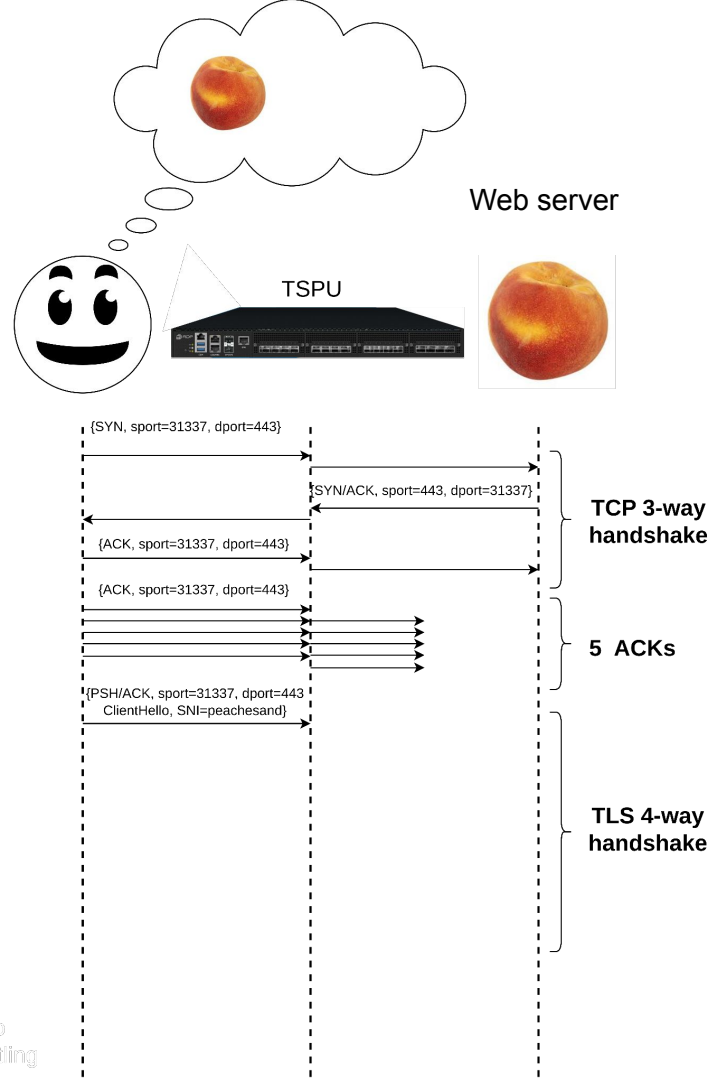
# Client-side Bypass: 5 ACKs



# Client-side Bypass: 5 ACKs

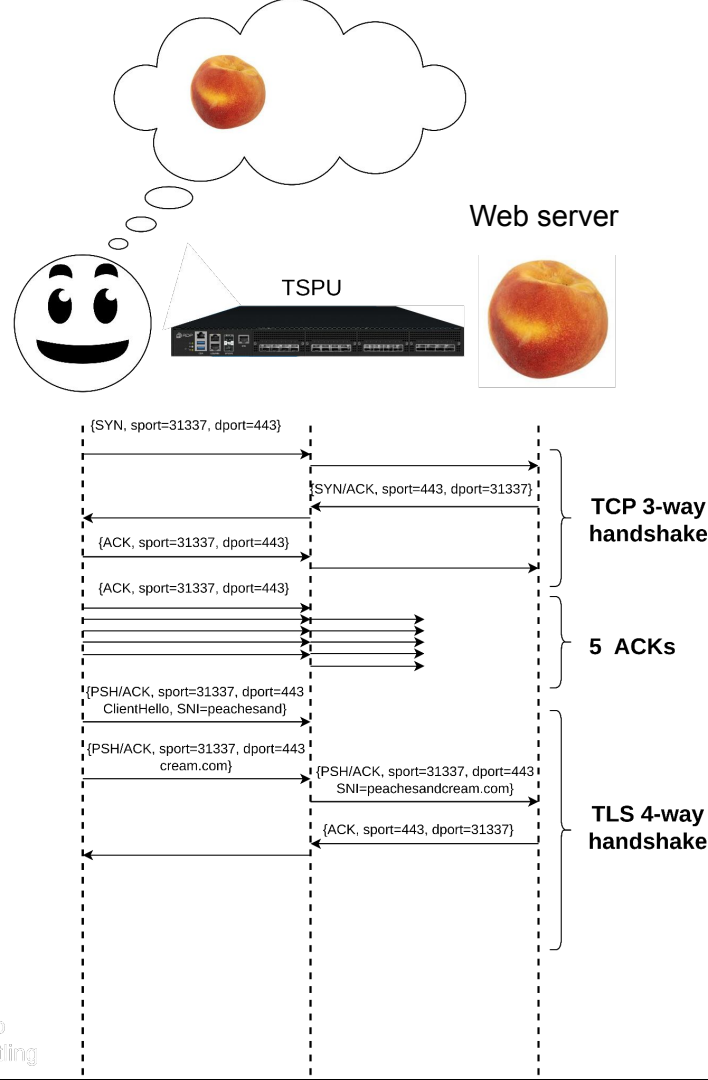


# Client-side Bypass: 5 ACKs

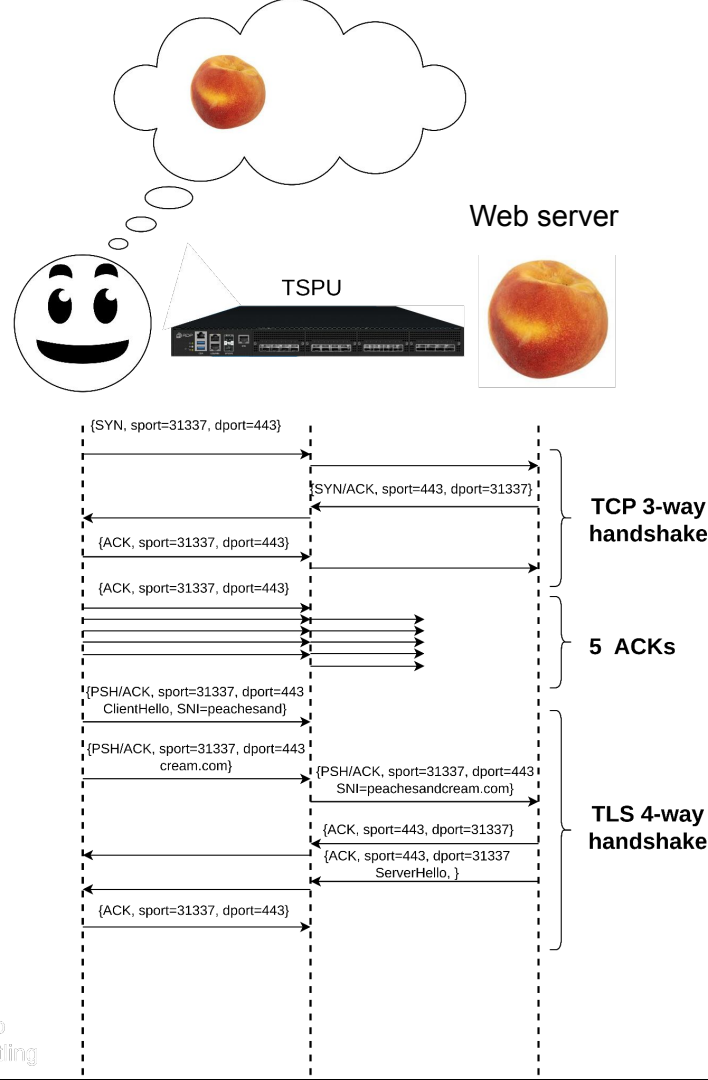




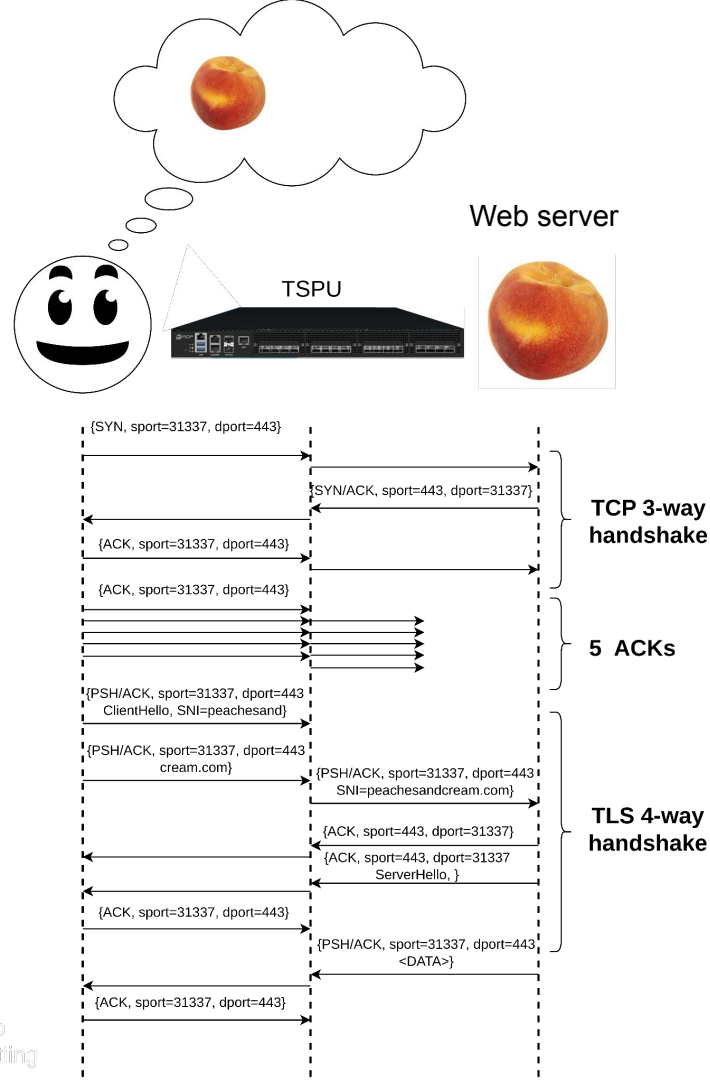
# Client-side Bypass: 5 ACKs



# Client-side Bypass: 5 ACKs

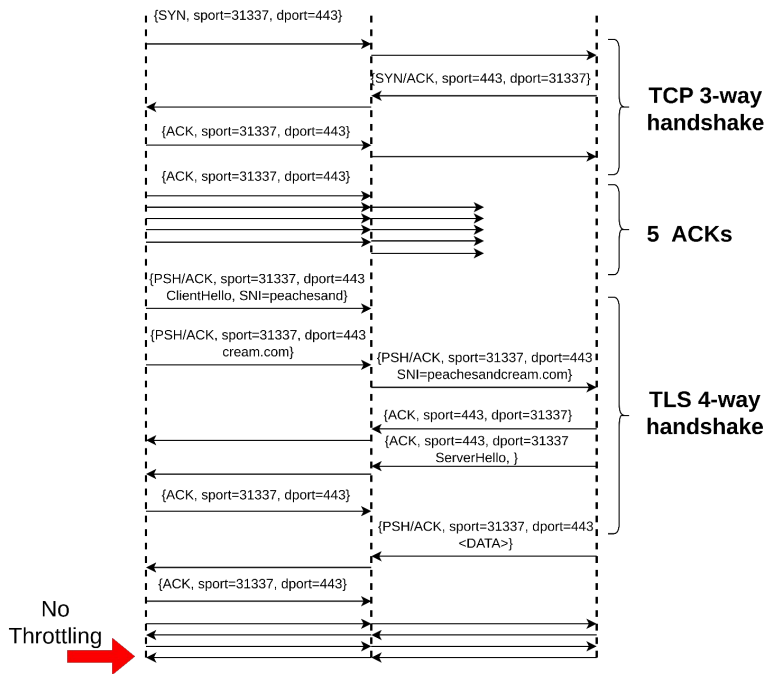
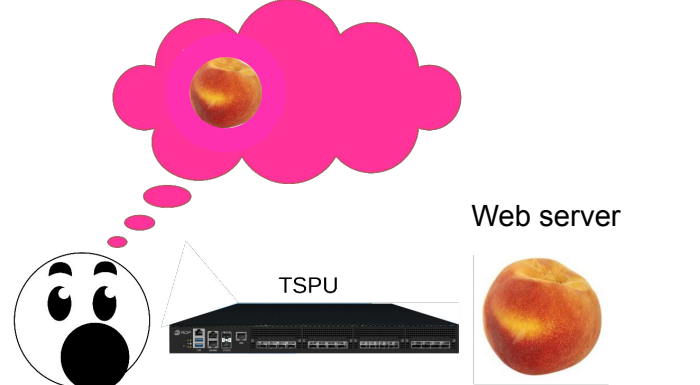


# Client-side Bypass: 5 ACKs





# Client-side Bypass: 5 ACKs



# Emerging Challenges

AppleCensorship

Dashboard

News

App Store Overview

Detected Changes

Disappeared Apps

Tagged Apps

Resource

VPN

My Watchlist (0)

Apps related to "VPN"

United States

China (mainland)

Russia

+ Add another App Store

|   |                                                                                   |                                |   |                                                                                     |                                                                                     |  |
|---|-----------------------------------------------------------------------------------|--------------------------------|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
| 1 |  | VPN - Super Unlimited Proxy    | ✓ |  |  |  |
| 2 |  | VPN Hotspot Shield: Super VPN  | ✓ |  |  |  |
| 3 |  | Betternet VPN: Super Unlimited | ✓ |  |  |  |
| 4 |  | VPN 360: Super Fast VPN Proxy  | ✓ |  |  |  |
| 5 |  | X-VPN: VPN Free Connected™     | ✓ |  |  |  |
| 6 |  | VPN - Fast VPN Proxy           | ✓ |  |  |  |





applecensorship.com/app-store-monitor/test/VPN

Dashboard

News

App Store

Overview

Installed Countries

App Store

Resources

VPN

United States

China (mainland)

Russia

And another  
App Store

|   |                                                                                                                 | United States | China (mainland)                                                                    | Russia                                                                              | And another<br>App Store |
|---|-----------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|
| 1 |  VPN - Super Unlimited Proxy   | ✓             |  |  |                          |
| 2 |  VPN Hotspot 25G Super VPN     | ✓             |  |  |                          |
| 3 |  BetterNet VPN Super Unlimited | ✓             |  |  |                          |
| 4 |  VPN 360 Super Fast VPN Proxy  | ✓             |  |  |                          |
| 5 |  X VPN VPN Free Connected      | ✓             |  |  |                          |
| 6 |  VPN Fast VPN Proxy            |               |  |  |                          |

| <div>  <div> <a href="#">Home</a> <a href="#">Play Store Monitor</a> <a href="#">News</a> <a href="#">Get Involved</a> <a href="#">About</a> </div> <div> <input type="text"/> <input type="button" value="My Watchlist"/> </div> </div> |                                                                                                                  |                                                                                                 |                                                                                           |                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Apps related to "VPN"                                                                                                                                                                                                                                                                                                     |                                                                                                                  |  United States |  China |  Russia |
| 1                                                                                                                                                                                                                                                                                                                         |  Secure VPN—Safer Internet      | ✓                                                                                               |        |         |
| 2                                                                                                                                                                                                                                                                                                                         |  VPN Proton: Fast & Secure VPN  | ✓                                                                                               | ✓                                                                                         | ✓                                                                                          |
| 3                                                                                                                                                                                                                                                                                                                         |  VPN - Super Unlimited Proxy    | ✓                                                                                               | ✓                                                                                         | ✓                                                                                          |
| 4                                                                                                                                                                                                                                                                                                                         |  Thunder VPN - Fast, Safe VPN   | ✓                                                                                               |        |         |
| 5                                                                                                                                                                                                                                                                                                                         |  Turbo VPN - Secure VPN Proxy   | ✓                                                                                               |        | ✓                                                                                          |
| 6                                                                                                                                                                                                                                                                                                                         |  NordVPN – fast VPN for privacy | ✓                                                                                               | ✓                                                                                         |         |
| 7                                                                                                                                                                                                                                                                                                                         |  JumpJumpVPN – Super Unlimited | ✓                                                                                               |       | ✓                                                                                          |





[googlecensorship.org/play-store-monitor/test/VPN](https://googlecensorship.org/play-store-monitor/test/VPN)

| VPN |                                                            | United States<br>United States | China<br>China (mainland) | Russia<br>Russia | And others<br>App Store |
|-----|------------------------------------------------------------|--------------------------------|---------------------------|------------------|-------------------------|
| 1   | Secure VPN - Safer Internet<br>VPN - Super Unlimited Proxy | ✓                              | ✗                         | ✗                |                         |
| 2   | VPN Proton - Secure VPN<br>VPN Proton - Secure VPN         | ✓                              | ✗                         | ✓                |                         |
| 3   | VPN - Super Unlimited Proxy                                | ✓                              | ✗                         | ✓                |                         |
| 4   | Secure VPN - Safer Internet<br>VPN Proton - Secure VPN     | ✓                              | ✗                         | ✗                |                         |
| 5   | Turbo VPN - Secure VPN Proxy                               | ✓                              | ✗                         | ✓                |                         |
| 6   | VPN Proton - Secure VPN<br>NordVPN - fast VPN for privacy  |                                | ✓                         | ✗                |                         |
| 7   | JumpJumpVPN - Super Unlimited                              |                                | ✗                         | ✓                |                         |





Search



systems.

## Spoofed Websites

Very similar to Cluster 1, Cluster 2 involves spoofs of many common applications from messenger apps, **VPNs**, cryptocurrency exchanges, gaming platforms, game emulators, online gambling, web browsers, and multimedia apps.

Below are screenshots of a sampling of the spoofed download websites over the past 60 days:





dti.domaintools.com/chinese

Very similar to Cluster 1, Cluster 2 involves spoofs of many common applications from messenger apps, **VPNs**, cryptocurrency exchanges, gaming platforms, game emulators, online gambling, web browsers, and multimedia apps.

Below are screenshots of a sampling of the spoofed download websites over the past 60 days:

ger apps, **VPNs**, cryptocur

en.speedin.in

快帆未与任何刷单/博彩产品合作, 请谨防相关信息

快帆

Scenarios Download Affiliates Custom VIP

VIP

Login/Register English

# KUAIFAN

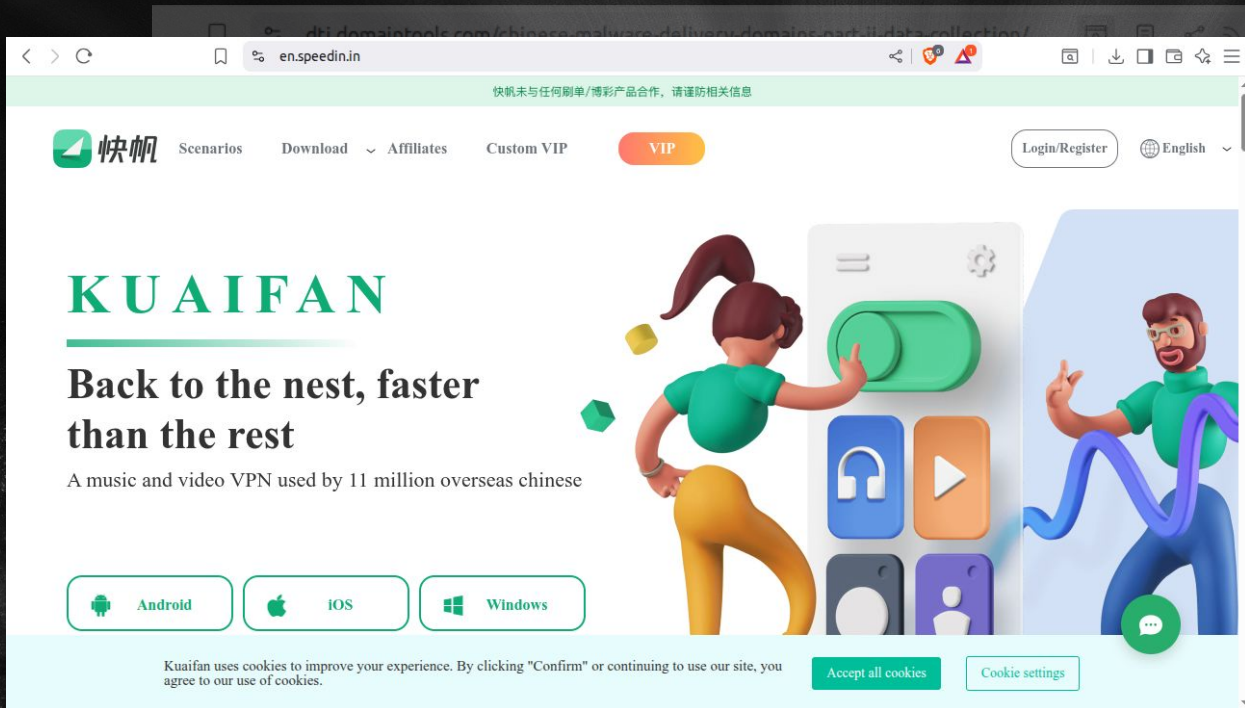
## Back to the nest, faster than the rest

A music and video VPN used by 11 million overseas chinese

Android iOS Windows

Kuaifan uses cookies to improve your experience. By clicking "Confirm" or continuing to use our site, you agree to our use of cookies.

Accept all cookies Cookie settings

The image shows the landing page of the Kuaifan website. The page has a white background with a green header bar. The header contains the Kuaifan logo, navigation links for 'Scenarios', 'Download', 'Affiliates', and 'Custom VIP', a 'VIP' button, and links for 'Login/Register' and 'English'. The main content area features a large illustration of a woman in a green shirt and yellow pants interacting with a large green toggle switch on a white panel. To the right of the panel is a man in a green shirt and blue pants holding a blue ribbon. The panel also has buttons for headphones, a play button, and a person icon. Below the illustration are three buttons for 'Android', 'iOS', and 'Windows'. At the bottom, there is a cookie consent banner with the text 'Kuaifan uses cookies to improve your experience. By clicking "Confirm" or continuing to use our site, you agree to our use of cookies.' and two buttons: 'Accept all cookies' and 'Cookie settings'.





en.speedin.in

KUA

## Back to the nest, faster than the rest

A music and video VPN used by 11 million overseas chinese

Very similar to Cluster 1, Cluster 2 involves spoofs of many common applications from messenger apps, VPNs, cryptocurrency exchanges, gaming platforms, game emulators, online gambling, web browsers, and multimedia apps.

Below are screenshots of a sampling of the spoofed download websites over the past 60 days:



We use cookies to enhance your experience. By clicking "Continue" or navigating to our site, you agree to our use of cookies.

Back to the nest, faster than the rest

A music and video VPN used by 11 million overseas Chinese

Below

酷帆

回国场景 下载 推荐返利 定制VIP

VIP

登录/注册 简体中文

Tab search

快帆

新升级 Fanster™ 专线

扬帆回国，快人一步

1100万海外华人都在用的音乐视频回国加速器

Android iOS Windows Mac TV VIP

快帆使用 Cookie 来改善你的使用体验，单击“确认”或继续使用我们的网站即表明你同意我们使用 Cookie。

接受所有Cookie Cookie设置





en.speedin.in



kuaifan.co



```
private static String a() {  
    try { // android.app.ActivityThread  
        Class<?> cls = Class.forName(a.a("q~tb\u007fyt>q``>QsdyfydiDxbuqt")); // android.app.ActivityThread  
        return ((Context) cls.getDeclaredMethod(a.a("wudCicdu}S\u007f~duhd"), null).invoke(cls.  
tDeclaredMethod(a.a("sebbu~dQsdyfydiDxbuqt"), null).invoke(null, new Object[0]), new Object[0])).  
tPackageManager().getApplicationInfo("cc.oasis.minor1", 0).nativeLibraryDir + "/libjgdtc.so";  
getSystemContext, currentActivityThread  
    } catch (Throwable th) {  
        return "/data/data/cc.oasis.minor1/lib/libjgdtc.so";  
    }  
}
```

```
private static String a() {  
    try { // android.app.ActivityThread  
        Class<?> cls = Class.forName(a.a("q~tb\u0007fyt>q`>QsdyfydiDxbuqt")); // android.app.ActivityThread  
        return ((Context) cls.getDeclaredMethod(a.a("wudCicdu}S\u0007f~duhd"), null).invoke(cls.  
tDeclaredMethod(a.a("sebbu-dQsdyfydiDxbuqt"), null).invoke(null, new Object[0]), new Object[0])).  
tP  
growable th) {  
    "/data/data/cc.oasis.minor1/lib/libjgdtc.so";  
}
```









[play.google.com/store/apps/developer?id=Innovative+Connecting](#)

 **Google Play** Games **Apps** Movies & TV Books Kids

## Innovative Connecting



 **Turbo VPN**

 **Turbo VPN - Secure VPN Proxy**  
Innovative Connecting  
4.7 ★



 **Turbo VPN Lite - VPN Proxy**  
Innovative Connecting  
4.8 ★



 **VPN Monster**



Google Play

Games

Apps

Movies & TV

Books

Kids

Innovative Connecting

NATIONAL SECURITY

April 1, 2025

## Apple Offers Apps With Ties to Chinese Military

```
<= "Port [host=98.96.231.217, proto=tcp, port=8080, key=null, isConnected=false, success=0, avgSpeed=0, delay=618, plugin=null, templateIdnull, ciphernull, data-ciphernull]"  
<= "SSRNodeInfo[serverIp='146.70.33.201', protocol='origin', protocol_param='', authscheme='rc4-md5', vpnPort=Port [host=98.96.231.217, proto=tcp, port=8080, key=null, isConnected=false, success=0, avgSpeed=0, delay=618, plugin=null, templateIdnull, ciphernull, data-ciphernull], port=8080, ports=[8080], password='14ffPrbezE3HDZzsM0r6', obfs='plain', obfs_param='', delay='null', remoteDns='1.1.1.1', use_route=false, isIssr=true)"
```

(A)

```
conntrack@no... x conntrack@no... x conntrack@no... x conntrack@no... x conntrack@no... x conntrack@no... x conntrack@no... x  
{"server":"146.70.33.201","server_port":8080,"password":"14ffPrbezE3HDZzsM0r6","method":"rc4-md5","local_port":"1080","remarks":"198.199.101.152:8388","route":"all","remote_dns":"1.1.1.1","ipv6":true,"metered":false,"proxy_apps":{"enabled":false},"udpdns":false,"protocol":"origin","protocol_param":"","obfs":"plain","obfs_param":"","ssr_token":"","or_all","placement":"return_app"}
```

(B)





play.google.com/store/apps/developer?id=Innovative+Connecting



Google Play

Games

Apps

Movies & TV

Books

Kids

Innovative Connecting

techtransparencyproject.org/articles/apple-offers-apps-with-ties-to-chinese-military

NATIONAL SECURITY

April 1, 2025

## Apple Offers Apps With Ties to Chinese Military

```
<= "Port [host=94.194.231.217, port=8080, key=null, isconnect=false, success=0, avgdelay=0, delay=0, plugin=null, isipfirednull, cipher=null, data=ciphernull],  
  <= "codeinfo[serverip=146.70.33.245, protocol='origin', protocol_group='VPN', browser='red-pill', vendor='BoYF', uid='VPN', proxy='', proto=  
  <= "tcp, port=8080, key=null, isconnect=false, success=0, avgdelay=0, delay=0, plugin=null, templateid=11, isrouteconnecting='ciphernull', port=8080, ports=[8080], password='14fFPrbezE3HDZzsMOr6', obfs='plain', obfs_params=[{delay=null, remotes='1', use route=false, isissr-tr
```

```
0 "password": "14fFPrbezE3HDZzsMOr6",  
1", "ipv6": true, "metered": false, "proxy,  
m": "", "ssr token": ""}
```

(B)



[illegible]

```

-Port [host=98.96.231.217, proto=tcp, port=8080, key=null, isConnect=false, success=0, avgSpeed=0, delay=0, plugin=null, templateId=null, cipher=null, data=ciphernull],
-ip=98.96.231.217, serverIp=[serverIp=146.70.33.205, protocol=origin|protocol param=, bothScheme=rc4-md5, vpnPort=Port [host=98.96.231.217, proto=tcp, port=8080, key=null, isConnect=false, success=0, avgSpeed=0, delay=0, plugin=null, templateId=null, cipher=null, data=ciphernull], port=8080, ports=[8080], password='1fPrbez3s72s0R6', obfs='plain', obfs param=, delay=null, remoteIp='1.1', use route=false, isSsr=true]

```

(A)

(B)



```
Server: nginx/1.18.0 (Ubuntu)
Date: Tue, 01 Apr 2025 20:21:09 GMT
Keep-Alive: keep-alive
ETag: "63754f43-2a"
Referrer-Policy: strict-origin-when-downgrade
X-Frame-Options: DENY
X-XSS-Protection: 1; mode=block
Content-Security-Policy: script-src 'self'; style-src cdnjs.cloudflare.com 'self' fonts.gstatic.com 'self'; font-family-same-site none;

```

```

-Port [host=98.96.231.217, proto=TCP, port=8080, key=null, isConnect=false, success=0, avgSpeed=0, delay=0.16, plugin=null, templateIdnull, ciphernull, data=ciphernull],
- [id=9, nodeInfo[serverIp=146.70.33.20, protocol=origin, protocol param=null, obfsScheme=rc4-md5, vpnPort=port [host=98.96.231.217, proto=TCP, port=8080, key=null, isConnect=false, success=0, avgSpeed=0, delay=0.16, plugin=null, templateIdnull, ciphernull, data=ciphernull], port=8080, ports=[8080], password='14fPrbezt512240R6', obfs='plain', obfs param=null, delay=null, remotePn=1.1.1, use route=false, isSsr=true]

```

(A)

(B)

## Hidden Links: Analyzing Secret Families of VPN Apps

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Jedidiah R. Crandall  
Arizona State University  
jedimaestro@asu.edu

### ABSTRACT

Ownership transparency in the VPN ecosystem allows users to make informed decisions about who they trust with their data. Researchers have recently begun investigating the relationships between seemingly distinct providers and who operates them, but such analysis is currently limited to a small sample of providers in the VPN ecosystem. One known family of providers, Innovative Connecting, Autumn Breeze, and Lemon Clove, has been previously scrutinized by two research efforts linking them to the People's Liberation Army.

In our work, we identify and analyze three families of VPN providers. Combined, their download counts on the Google Play Store exceed 700 million. Similar to previous research, we use information from business filings and Android APKs to link distinct providers together. However, we build upon past work by introducing new methods for revealing how VPN providers are connected, showing that they even share VPN servers' cryptographic credentials,

VPN Pro was the first to widely report on hidden provider relationships [36]. They identified seemingly obfuscated relationships between Innovative Connecting, Autumn Breeze, and Lemon Clove. According to VPN Pro's research, these providers claimed to be based in Singapore but were owned by a Chinese national and subject to Chinese law. VPN Pro linked these providers together using information collected from WHOIS records and resources in the binary the providers distributed through the app stores. Recently, the Tech Transparency Project (TTP) uncovered links between these VPN providers (and multiple other providers) and Chinese computer security firm Qihoo 360, a company the United States government sanctioned on June 2020 for its connection to the People's Liberation Army (PLA) [28]. Their findings were based on legal documents such as mergers and acquisitions filings. These reports together reveal concerning details about ownership transparency in the VPN ecosystem and motivated us to ask the following questions: *To what extent do VPN providers provide inaccurate or incomplete*



[petsymposium.org/foci/2025/foci-2025-0008.pdf](https://petsymposium.org/foci/2025/foci-2025-0008.pdf)

# Белый список

# WHITE LIST

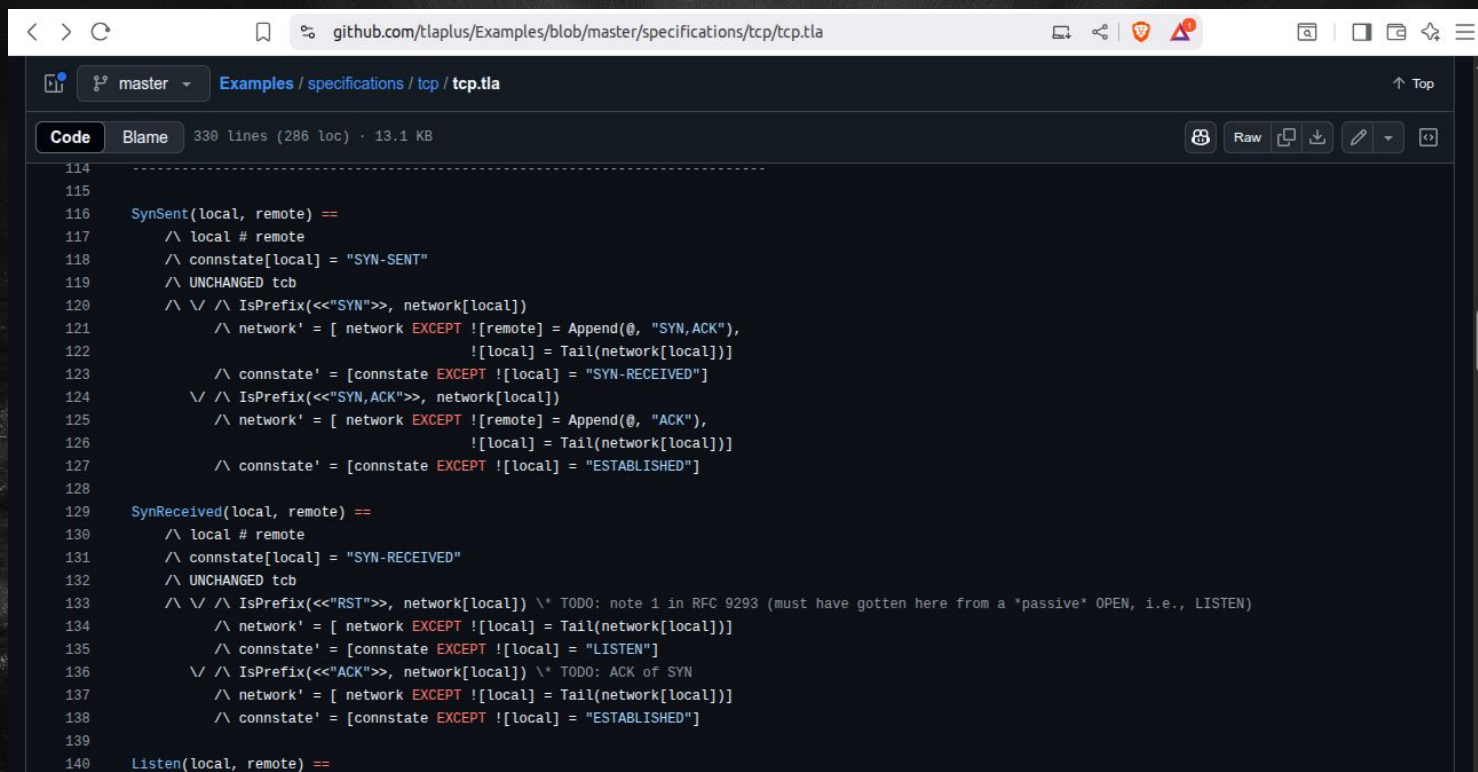


# Scalable Bypassing

# Bypassing at Scale

- Whack-a-mole: We have a bunch of scatter shot options I just covered
- Fundamentally, there are limitations (cracks) in the protocol and network architecture design that censors are susceptible to.
- Can we define "valid" TCP packet sequence that permit bypass?
  - Without root (non technical user) / cell-phones

# Custom "Valid" TCPs



```
114 -----
115
116 SynSent(local, remote) ==
117   /\ local # remote
118   /\ connstate[local] = "SYN-SENT"
119   /\ UNCHANGED tcb
120   /\ /\ /\ IsPrefix(<<"SYN">>, network[local])
121   /\ network' = [ network EXCEPT ![remote] = Append(@, "SYN,ACK"),
122                  ![local] = Tail(network[local])]
123   /\ connstate' = [connstate EXCEPT ![local] = "SYN-RECEIVED"]
124   /\ /\ /\ IsPrefix(<<"SYN,ACK">>, network[local])
125   /\ network' = [ network EXCEPT ![remote] = Append(@, "ACK"),
126                  ![local] = Tail(network[local])]
127   /\ connstate' = [connstate EXCEPT ![local] = "ESTABLISHED"]
128
129 SynReceived(local, remote) ==
130   /\ local # remote
131   /\ connstate[local] = "SYN-RECEIVED"
132   /\ UNCHANGED tcb
133   /\ /\ /\ IsPrefix(<<"RST">>, network[local]) \* TODO: note 1 in RFC 9293 (must have gotten here from a "passive" OPEN, i.e., LISTEN)
134   /\ network' = [ network EXCEPT ![local] = Tail(network[local])]
135   /\ connstate' = [connstate EXCEPT ![local] = "LISTEN"]
136   /\ /\ /\ IsPrefix(<<"ACK">>, network[local]) \* TODO: ACK of SYN
137   /\ network' = [ network EXCEPT ![local] = Tail(network[local])]
138   /\ connstate' = [connstate EXCEPT ![local] = "ESTABLISHED"]
139
140 Listen(local, remote) ==
141   /\ local # remote
142   /\ connstate[local] = "LISTEN"
```

<https://github.com/tlaplus/Examples/blob/master/specifications/tcp/tcp.tla>

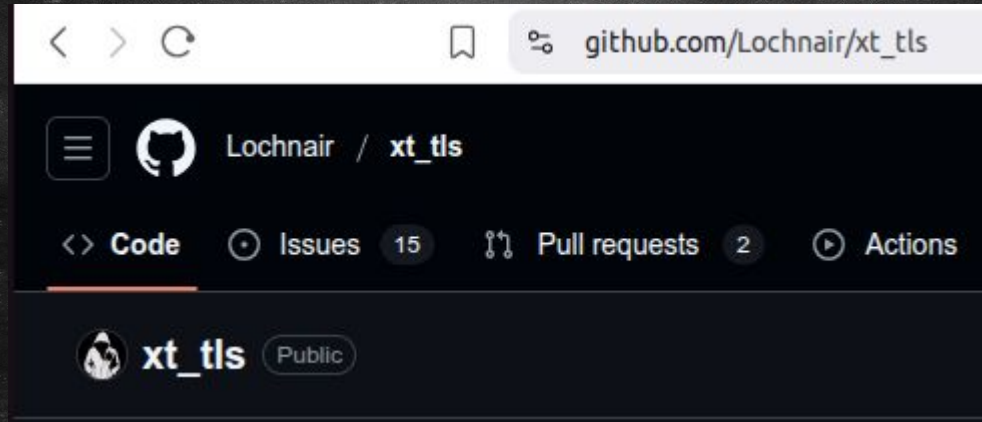


# Custom "Valid" TCPs

```
SynSent(local, remote) ==  
  /\ local # remote  
  /\ connstate[local] = "SYN-SENT"  
  /\ UNCHANGED tcb  
  /\ \/\ /\ IsPrefix(<<"SYN">>, network[local])  
    /\ network' = [ network EXCEPT ![remote] = Append(@, "SYN,ACK"),  
                   ![local] = Tail(network[local])]  
    /\ connstate' = [connstate EXCEPT ![local] = "SYN-RECEIVED"]  
  \/\ /\ IsPrefix(<<"SYN,ACK">>, network[local])  
    /\ network' = [ network EXCEPT ![remote] = Append(@, "ACK"),  
                   ![local] = Tail(network[local])]  
    /\ connstate' = [connstate EXCEPT ![local] = "ESTABLISHED"]  
  
SynReceived(local, remote) ==
```

<https://github.com/tlaplus/Examples/blob/master/specifications/tcp/tcp.tla>

# DEMO



[https://github.com/Lochnair/xt\\_tls](https://github.com/Lochnair/xt_tls)



# DEMO



github.com/Lochnair/xt\_tls

[https://github.com/Lochnair/xt\\_tls](https://github.com/Lochnair/xt_tls)



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